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Mycobacterium tuberculosis PhoP integrates stress response to intracellular survival by maintenance of cAMP level

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Abstract

Survival of *M. tuberculosis* within the host macrophages requires the virulence regulator PhoP, but the underlying reason remains unknown. cAMP is one of the most widely used second messengers, which impacts on a wide range of cellular responses in microbial pathogens including *M. tuberculosis*. Herein, we hypothesized that intra-mycobacterial cAMP level could be controlled by the *phoP* locus since the major regulator plays a key role in bacterial responses against numerous stress conditions. A transcriptomic analysis reveals that PhoP functions as a represtsor of cAMP-specific phosphodiesterase (PDE) Rv0805, which hydrolytically degrades cAMP. In keeping with these results, we find specific recruitment of the regulator within the promoter region of *rv0805* PDE, and absence of *phoP* or ectopic expression of *rv0805* independently accounts for elevated PDE synthesis leading to depletion of intra-mycobacterial cAMP level. Thus, genetic manipulation to inactivate PhoP-*rv0805*-cAMP pathway decreases cAMP level, stress tolerance and intracellular survival of the bacilli.

eLife assessment

This **important** study describes how PhoP regulates cyclic-AMP production in the human pathogen Mycobacterium tuberculosis. The authors provide **convincing** evidence that PhoP acts as a repressor of the cyclic-AMP-specific phosphodiesterase, Rv0805, which can degrade cyclic-AMP. The work will be of interest to bacteriologists and whilst the revised manuscript has been substantively improved, there are some outstanding points that could improve clarity and presentation.

Introduction

M. tuberculosis, the causative agent of pulmonary tuberculosis, encounters diverse environmental conditions during infection, persistence and transmission of the disease (Ehrt *et al*, 2018; Ernst, 2012; Russell, 2011). However, the intracellular pathogen is exceptionally capable to adjust and survive within diverse intracellular environments. Mycobacterial adaptive response to different stages of infection is achieved via fine tuning of regulation of gene expression using an extensive repertoire of more than 100 transcriptional regulators, 11 two-component systems, 6-serine-threonine protein kinases and 13 alternative sigma factors suggesting that a very complex transcriptional program is important for *M. tuberculosis* pathogenesis. While regulation of gene expression as a consequence of interaction of tubercle bacilli with its immediate environment remains critically important (Rohde *et al*, 2007), defining these regulatory pathways represent a major challenge in the field.

3', 5'-cyclic adenosine monophosphate (cAMP), one of the most widely used second messengers, impacts on a wide range of cellular responses in microbial pathogens including *M. tuberculosis* (McDonough & Rodriguez, 2011). The bacterial genome encodes at least 15 adenylate cyclases, including one of the adenylate cyclases (Rv0386) which is required for virulence (Agarwal *et al*, 2009), and multiple cAMP regulated effector proteins(Banerjee *et al*, 2015; Johnson & McDonough, 2018; McDonough & Rodriguez, 2011; Shenoy & Visweswariah, 2006). cAMP levels are elevated upon infection of macrophages by pathogenic mycobacterium (Bai *et al*, 2009) and addition of exogenous cAMP was shown to influence mycobacterial protein expression (Gazdik & McDonough, 2005). While intra-bacterial cAMP responds to macrophage environment and carries out specific functions like gene expression and protein function under host-related conditions (Bai *et al*, 2011; Bai *et al*., 2009; Johnson *et al*, 2017; Johnson & McDonough, 2018; Knapp & McDonough, 2014), secreted cAMP controls macrophage signalling (Agarwal *et al*., 2009; Agarwal *et al*, 2006; Gazdik *et al*, 2009; Johnson *et al*., 2017; Nambi *et al*, 2013; Ranganathan *et al*, 2016) (Rittershaus *et al*, 2018). cAMP signalling remains essential to *M. tuberculosis* pathogenesis. Agarwal and colleagues had shown that burst in synthesis of cAMP upon infection of macrophages, improved bacterial survival by interfering with host signalling pathways (Agarwal *et al*., 2009). In keeping with this, anti-tubercular compounds that interfere with mycobacterial cAMP levels also impact intracellular growth of mycobacteria (Bai *et al*., 2011; Johnson *et al*., 2017; Knapp *et al*, 2015; Nambi *et al*., 2013; Rittershaus *et al*., 2018; VanderVen *et al*, 2015; Wilburn *et al*, 2022). Together, these results underscore the significance of cAMP signalling in mycobacteria.

cAMP signalling is controlled at the transcriptional level. Previous *in silico* studies had identified two out of 10 predicted nucleotide binding proteins of *M. tuberculosis* as members of the CRP/FNR superfamily of transcriptional regulators (McCue *et al*, 2000). These are CRP (cyclic AMP Receptor protein), encoded by *rv3676* and CMR (cAMP macrophage regulator), encoded by *rv1675c*, respectively. Of these two, CRP becomes activated upon cAMP binding, and functions as a global regulator of ~100 genes (Agarwal *et al*., 2006; Bai *et al*, 2005; Rickman *et al*, 2005). Consequently, deletion of the *crp* locus significantly impairs mycobacterial growth and attenuates virulence of the bacilli in a mouse model (Rickman *et al*., 2005). In contrast, CMR is necessary for regulated expression of genes involved in virulence and persistence including members of the dormancy regulon (Gazdik *et al*., 2009; Ranganathan *et al*., 2016; Smith *et al*, 2017). Together, these results strongly suggest that a balance between cAMP synthesis by adenylate cyclases and cAMP degradation by phosphodiesterases contributes to rapid adaptive response of mycobacteria in a hostile intracellular environment (Johnson & McDonough, 2018; McDonough & Rodriguez, 2011). However, very little is known about the underlying mechanism of mycobacterial cAMP homeostasis.

M. tuberculosis encounters a hostile environment within the host. Among the hostile conditions is the acidic pH stress and exposure to host immune effectors such as NO (Nathan & Shiloh, 2000; Rustad *et al*, 2009; Wayne & Sohaskey, 2001). A growing body of evidence connects virulence-associated mycobacterial *phoP* locus with varying environmental conditions, including acid stress (Abramovitch *et al*, 2011; Bansal *et al*, 2017; Tan *et al*, 2013), heat-shock (Sevalkar *et al*, 2019; Singh *et al*, 2014) and integration of acid stress response to redox homeostasis (Baker *et al*, 2019; Baker *et al*, 2014; Goar *et al*, 2022). Disruption of *phoP*, the gene encoding the response regulator of the PhoPR two-component signal transduction system (Gupta *et al*, 2006) significantly reduces *in vivo* multiplication of the bacilli (Walters *et al*, 2006), suggesting that PhoPR remains essential for virulence (Perez *et al*, 2001). Moreover, the mutant lacking this system shows a significantly lowered synthesis of cell-wall components diacyltrehaloses, polyacyltrehaloses, and sulfolipids, specific to pathogenic mycobacterial species (Gonzalo Asensio *et al*, 2006a; Goyal *et al*, 2011; Walters *et al*., 2006). In fact, the significant attenuation of *phoPR* deletion strain forms the basis of the mutant being considered in trials as a vaccine strain (Arbues *et al*, 2013). Recent transcriptomic analyses revealed that approximately 2% of the H37Rv genome is regulated by PhoP (Solans *et al*, 2014; Walters *et al*., 2006), and mycobacterial gene expression in response to acidic pH significantly overlaps with the PhoP regulon (Rohde *et al*., 2007). Consistent with this, a large subset of PhoPR-regulated low-pH-inducible genes are induced immediately following *M. tuberculosis* phagocytosis and remain induced during macrophage infection (Gonzalo Asensio *et al*, 2006b; Martin *et al*, 2006; Perez *et al*., 2001; Walters *et al*., 2006). Additional evidences coupled with more recent results suggest that during onset of macrophage infection, PhoPR activation is linked to acidic pH and the available carbon source, suggesting a physiological link between pH, carbon source, and macrophage pathogenesis (Baker *et al*., 2019).

In this study, we hypothesized that intra-mycobacterial cAMP level could be determined by the *phoP* locus since the major regulator has been implicated in regulation of bacterial responses against numerous stress conditions, many of which function as signals to activate cAMP synthesizing diverse adenylate cyclases (Knapp & McDonough, 2014). Our results connect virulence regulator PhoP with intra-mycobacterial cAMP level. We discovered that PhoP regulates expression of cAMP-specific phosphodiesterase *rv0805*, which hydrolytically degrades cAMP. To further probe the regulation, we demonstrate that under the condition which activates PhoP-PhoR system, PhoP in a PhoR-dependent mechanism represses transcription of *rv0805* through direct DNA binding at the upstream regulatory region. These observations account for a consistently lower level of cAMP in a PhoPR-deleted *M. tuberculosis* H37Rv (*phoPR*-KO) relative to the WT bacilli, and establishes the molecular mechanism of regulation of cAMP level, absence of which strikingly impacts phagosome maturation, and reduces mycobacterial survival within macrophages and mice. Together, the newly identified mechanism of regulation of cAMP level allows intra-phagosomal survival and growth program of mycobacteria.

Results

Intra-mycobacterial cAMP level is regulated by the *phoP* locus

We compared cAMP levels of WT and *phoPR*-KO (mutant lacking both copies of *phoP* and *phoR*), grown under normal, NO stress and acid stress conditions (**Fig. 1A**). *phoPR*-KO showed a significantly lower level of cAMP relative to the WT bacilli, both under normal and stress conditions. Complementation of the mutant (Compl.) could restore cAMP to the WT level. A higher cAMP level in the complemented strain under NO stress is possibly attributable to reproducibly higher *phoP* expression in the complemented mutant under specific stress conditions (Khan et al., 2022). Because bacterial growth often varies under stress conditions, and growth inhibition can influence cAMP level, we compared viability of mycobacterial strains under normal and indicated stress conditions (conditions of cAMP measurements) by determining the bacterial CFU (Figure 1 -

figure supplement 1). Note that for *in vitro* viability under specific stress conditions, indicated mycobacterial strains were grown to mid log phase (OD_{600} 0.4-0.6) and exposed to acidic media (7H9 media, pH 4.5(Gouzy *et al*, 2021) for further two hours at 37°C. Likewise, for NO stress, cells grown to mid-log phase were exposed to 0.5 mM DetaNonoate for 40 minutes. Our results suggest that WT and *phoPR*-KO under carefully controlled stress conditions display comparable viability, indicating that variation in viable cell counts of the mutant under specific stress conditions do not account for lower cAMP level. From these results, we conclude that PhoP plays a major role in maintaining intra-mycobacterial cAMP level.

To investigate the possibility that *phoPR*-KO secretes out more cAMP and therefore, shows a lower cytoplasmic level, we compared cAMP secretion of WT and the mutant (**Fig. 1B**). The WT and mutant strains were grown as described in the Methods and culture filtrates (CF) as well as cell lysates (CL) were collected as described previously (Anil Kumar *et al*, 2016). Our results demonstrate that the mutant reproducibly secretes lower amount of cAMP relative to the WT bacilli, and cAMP secretion is fully restored in the complemented mutant (Compl.). As the fold difference of secretion (~ 1.25-fold) is not much different relative to the fold difference in intra-mycobacterial cAMP level (~2-fold), we suggest that lower cAMP level of the mutant is not due to its higher efficacy of cAMP secretion. **Fig. 1C** confirms absence of autolysis of mycobacterial cells as GroEL2 was undetectable in the culture filtrates (CF).

PhoP functions as a repressor of *rv0805*

We next investigated role of the *phoP* locus on expression of mycobacterial adenylate cyclases (ACs) and phosphodiesterases (PDEs), which synthesize and degrade cAMP, respectively (**Fig. 2A**). The selection of ACs and PDEs were based on two key points. First, we have chosen ACs which are activated by known signals (Knapp & McDonough, 2014). Second, we reasoned that the previously reported ACs were activated under environmental conditions, which are linked to mycobacterial *phoP* locus (Bansal *et al*., 2017; Goar *et al*., 2022). Our RT-qPCR results using gene-specific primer pairs (Supplementary file 1a) suggest that expression of adenylate cyclases including *rv0386*, *rv1264*, *rv1647* and *rv2488c* (Agarwal *et al*., 2009; Dass *et al*, 2008; Dittrich *et al*, 2006; Knapp & McDonough, 2014) do not appear to be regulated by the *phoP* locus. However, a significant activation of adenylate cyclase *rv0891c* (4±0.05-fold), and repression of phosphodiesterase *rv0805* expression (6.5±0.7-fold), respectively, were dependent on the *phoP* locus. Although Rv0891c was suggested as one of the *M. tuberculosis* H37Rv adenylate cyclases, the protein lacks most of the important residues conserved for adenylate cyclase family of proteins (Zaveri *et al*, 2021). On the other hand, *rv0805* encodes for a cAMP specific PDE, present only in slow growing pathogenic *M. tuberculosis* (Matange *et al*, 2013; Shenoy *et al*, 2007; Shenoy *et al*, 2005). Although expression of *rv0805* was restored at the level of WT in the complemented mutant (Compl.), we observed poor restoration of expression of *rv0891c* in the Compl. strain. Further, expression of PDE *rv1339* which contributes to mycobacterial cAMP level (Thomson *et al*, 2022) remains unaffected by the *phoP* locus. Therefore, we focussed our attention on the biological significance of PhoP-dependent regulation of *rv0805*. It should be noted that expressions of *rv1357c* and *rv2837c*, encoding PDEs for cyclic di-GMP (Flores-Valdez *et al*, 2015) and cyclic di-AMP (Valadares & Woo, 2017), respectively, remained unchanged in *phoPR*-KO.

As we reproducibly observed activation of *rv0805* expression in *phoPR*-KO (relative to WT), we investigated whether acidic pH conditions under which *phoPR* system is activated (Abramovitch *et al*., 2011; Bansal *et al*., 2017), impacts expression of *rv0805* in WT bacilli (**Fig. 2B**). Our results show that repression of *rv0805* is significantly higher in WT grown under acidic conditions (pH 4.5) relative to normal conditions (pH 7.0) of growth. This observation is consistent with RNA-seq data displaying significant down-regulation of *rv0805* in WT bacilli grown under acidic pH conditions relative to normal conditions of growth (GEO Accession number: GSE180161). As a control, expression of PDE *rv1339* which also degrades cAMP, remains unaffected under acidic conditions of growth. The finding that acidic pH (pH4.5) conditions of growth promoted PhoP-dependent repression of *rv0805*, prompted us to investigate whether PhoP directly binds to *rv0805*

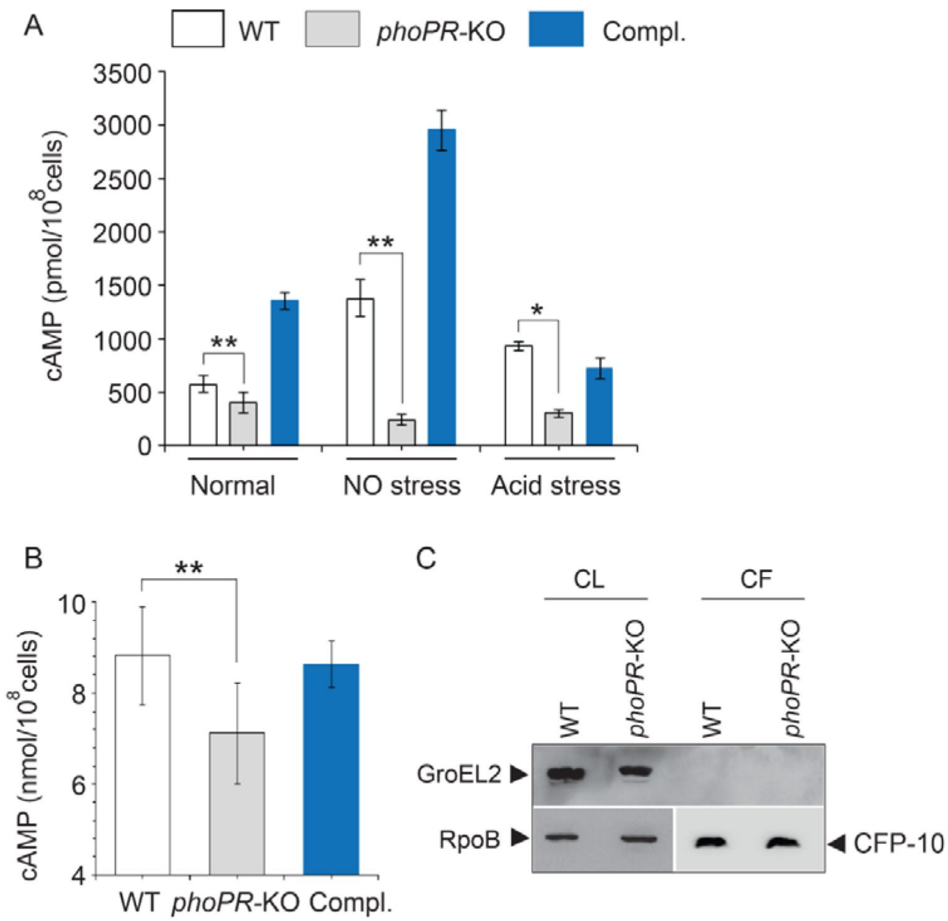

Figure 1

PhoP contributes to maintenance of mycobacterial cAMP level.

(A) Intra-mycobacterial cAMP levels were determined by a fluorescence-based assay as described in the Methods, and compared for indicated mycobacterial strains, grown under normal or specific stress conditions. For acid stress, mycobacterial strains were initially grown to mid-log phase (OD_{600} 0.4 to 0.6), and then it was transferred to acidic pH (7H9 media, pH 4.5) for further 2 hours of growth at 37°C. For NO stress, cells grown to mid-log phase were exposed to 0.5 mM DataNonoate for 40 minutes. The data represent average values from three biological repeats (*P≤0.05, **P≤0.01). (B) To compare secretion of cAMP by WT and *phoPR*-KO, cAMP levels were also determined in the corresponding culture filtrates (CF). (C) Immunoblotting analysis of 10 µg of cell lysates (CL) and 20 µg of culture filtrates (CF) of indicated *M. tuberculosis* strains. α-GroEL2 was used as a control to verify cytolysis of cells, CFP-10 detected as a secreted mycobacterial protein in the culture filtrates, and RpoB used as the loading control.

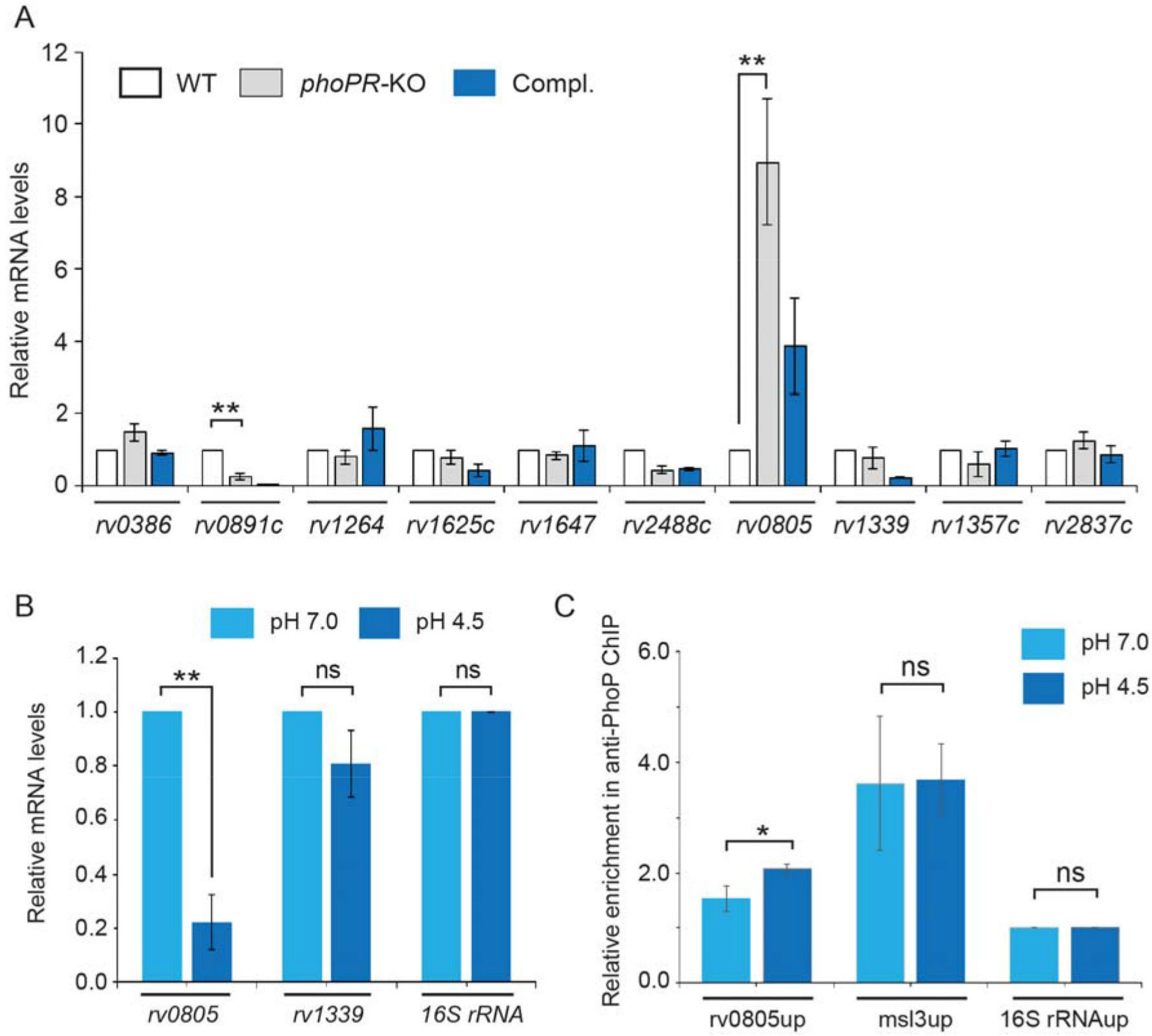

Figure 2

PhoP regulates expression of phosphodiesterase (PDE) *rv0805*.

(A) To investigate regulation of cAMP level, mRNA levels of well-characterized adenylate cyclases, and PDEs were compared in indicated mycobacterial strains by RT-qPCR as described in the Methods. The results show average values from biological triplicates, each with two technical repeats. Note that the difference in expression levels of *rv0805* between WT and *phoPR*-KO was significant (p<0.01), whereas the fold difference in mRNA level between WT and the complemented mutant (Compl.) remains nonsignificant (not indicated). (B) To determine the effect of acidic pH conditions of growth, mycobacterial *rv0805* expression was compared in WT grown under normal (pH 7.0) and acidic pH (pH 4.5). Average fold difference in mRNA levels from biological duplicates (each with a technical repeat) were measured as described in the Methods (**P≤0.01). As controls, expression of *rv1339* and *16S rDNA* were also measured. Non-significant difference is not indicated. (C) *In vivo* PhoP binding to *rv0805* promoter (rv0805up) was compared in WT grown under normal and acidic conditions of growth using anti-PhoP antibody followed by ChIP-qPCR. Fold enrichment data represent mean values of two independent experiments with a statistically significant fold difference (***P*-value <0.01; **P*-value <0.05). The upstream regulatory regions of 16S rRNA (16S rRNAup) and msl3 (msl3up) were used as negative and positive controls, respectively. The assay conditions, sample analyses, and detection are described in the Methods section.

promoter. To this end, we next examined *in vivo* recruitment of PhoP within the *rv0805* promoter by chromatin immunoprecipitation (ChIP) experiments (**Fig. 2C**). In this assay, formaldehyde-cross-linked DNA-protein complexes of growing *M. tuberculosis* cells were sheared to generate fragments of average size ≍500 bp. Next, ChIP experiments utilized anti-PhoP antibody, and IP DNA was analysed by real-time qPCR relative to a mock sample (without antibody as a control) using FPrv0805up/RPrv0805up as the primer pair (Supplementary file 1a). Our results show that under normal condition (light blue bars) rv0805up showed an insignificant enrichment of PCR signal for PhoP relative to mock (no antibody control) sample, suggesting low affinity DNA binding of PhoP under normal conditions. However, IP samples from cells grown under acidic pH showed a significantly higher enrichment of PhoP at the *rv0805* promoter (rv0805up; compare *light blue* bars with *dark blue* bars). As controls, promoter of *msl3* (msl3up) which is controlled by PhoP, and non-specific 16S rRNAup showed comparable enrichment, and no enrichment under normal and acidic conditions, respectively. Thus, ChIP data showing PhoP recruitment under acidic pH conditions is in agreement with low pH-specific impact of PhoP on *rv0805* expression (**Fig. 2B**). Note that PhoP binding to msl3up was used as a positive control.

To examine DNA binding *in vitro*, we first probed for the PhoP binding site within the upstream regulatory region of *rv0805* (rv0805up) by MEME Bioinformatic software using the consensus PhoP binding sequence (He & Wang, 2014). Our results suggest that the likely PhoP binding sequence spans from −127 to −110 (relative to the ORF start site) of rv0805up (Figure 2-figure supplement 1A) (p=0.000726). Next, recombinant PhoP was phosphorylated by acetyl-phosphate (AcP) and used in EMSA experiments as described earlier (Pathak *et al*, 2010). Consistent with the presence of a PhoP binding site, EMSA results demonstrate that P∼PhoP binds to radio-labelled rv0805up to form a complex stable to gel electrophoresis (Figure 2-figure supplement 1B).

Probing PhoP-dependent regulation of *M. tuberculosis rv0805*

To examine whether the regulatory effect was attributable to PhoP activation via phosphorylation, we next grew *phoPR*-KO complemented with either *phoP* (**Fig. 3A**) or the entire *phoPR* encoding ORFs (**Fig. 3B**), both under normal (pH 7.0; empty bars) and acidic (pH 4.5; black bars) conditions and compared relative expression of *rv0805*. Although both strains expressed *phoP*, the former strain lacked a functional copy of *phoR*, the cognate sensor kinase which phosphorylates PhoP (Gupta *et al*., 2006). *M. tuberculosis* H37Rv lacking a *phoR* gene (*phoPR*-KO*::phoP*) did not show a low pH-dependent repression of *rv0805* expression. However, similar to WT bacilli, we observed a low pH-dependent significant down-regulation of *rv0805* expression in *phoPR*-KO*::phoPR* (Compl.). Note that a comparable expression of PDE *rv1339* was observed in both strains regardless of growth conditions. These results indicate that acidic pH-dependent repression of *rv0805* expression *in vivo* is attributable to P∼PhoP requiring the presence of PhoR.

To examine the effect of Rv0805 on mycobacterial cAMP level, we next expressed a copy of *rv0805* in WT bacteria (referred to as WT-Rv0805) (**Fig. 3C**). *rv0805* ORF was cloned within the multicloning site of pSTki (Parikh *et al*, 2013) between the EcoRI and HindIII sites under the control of P_{myc1}*tetO* promoter, and expression of *rv0805* under non-inducing condition was verified by determining the mRNA level (Figure 3 - figure supplement 1A). Although copy number of episomal vectors with pAl5000 origin of replication (*oriM*) have been reported to be 3 by Southern hybridization (Ranes *et al*, 1990), in this case wild-type and mutant Rv0805 proteins were expressed from single-copy chromosomal integrants (Parikh *et al*., 2013). We then assessed the impact of Rv0805 on intra-mycobacterial cAMP level (**Fig. 3C**). Consistent with a previous study (Agarwal *et al*., 2009), WT-Rv0805 showed a significant depletion (2.1± 0.7-fold) of intra-mycobacterial cAMP relative to WT bacteria. To confirm that the reduced level of mycobacterial cAMP is attributable to *rv0805* expression, we also expressed *rv0805M*, a mutant Rv0805 lacking phosphodiesterase activity. As structural data coupled with biochemical evidences reveal that Asn-97 at the enzyme active site plays a key role in phosphodiesterase activity of Rv0805 (Shenoy *et al*., 2007; Shenoy *et al*., 2005), the mutant Rv0805M was constructed by changing the conserved Asn-97 to Ala. WT-Rv0805M showed an insignificant variation of cAMP level relative to WT,

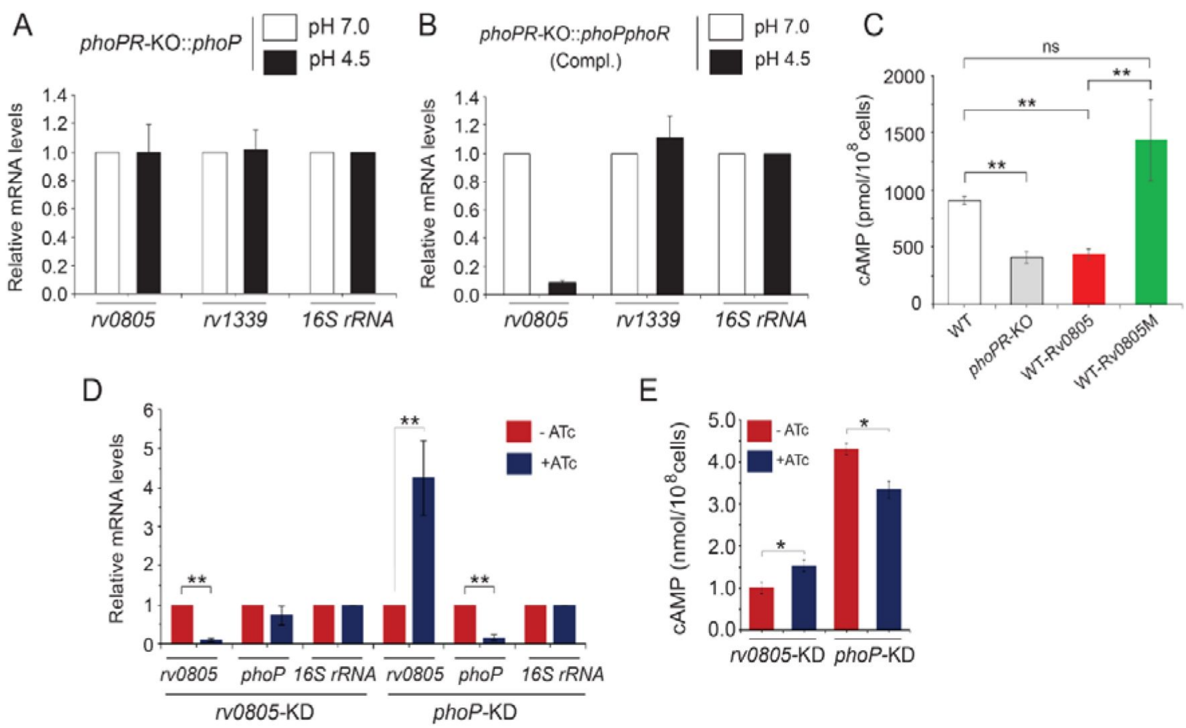

Figure 3.

PhoP dependent repression of *rv0805* to maintain mycobacterial cAMP level requires the presence of PhoR.

(A-B) To determine the impact of PhoR (the cognate sensor kinase of PhoP), expression of *rv0805* was compared in indicated *M. tuberculosis* H37Rv strains: (A) *phoPR*-KO::*phoP* (*phoPR* mutant complemented with *phoP*) and (B) *phoPR*-KO::*phoPR* (*phoPR* mutant complemented with *phoP-phoR*) under normal and acidic conditions of growth. As expected, *phoPR*-KO::*phoPR* (Compl.) shows a significant repression of *rv0805* (but not *rv1339*) under acidic pH (***P<0.001). However, *rv0805* expression remains comparable in *phoPR*-KO::*phoP* under normal as well as acidic conditions of growth. As a control, *rv1339* fails to show a variable expression in indicated mycobacterial strains. (C) To determine the effect of ectopic expression of *rv0805* on intra-mycobacterial cAMP level, WT and mutant Rv0805 proteins (Rv0805M, defective for phosphodiesterase activity) were expressed in *M. tuberculosis* H37Rv (to construct WT-Rv0805, and WT-Rv0805M, respectively) as described in the Methods section. Similar to *phoPR*-KO, WT-Rv0805 (but not WT-Rv0805M) showed a considerably lower level of cAMP relative to WT bacteria. Significance in variation of cAMP levels were determined by paired student's t-test (**P<0.01). (D-E) Relative expression of *phoP* and PDE in *phoP*-KD and *rv0805*-KD (*phoP* and *rv0805* knock-down constructs, respectively). In keeping with elevated expression of *rv0805* in *phoPR*-KO, *phoP*-KD shows an elevated expression of *rv0805* relative to WT bacilli. In contrast, *phoP* expression level remains unaffected in *rv0805*-KD mutant. Panel E measured corresponding intra-bacterial cAMP levels in the respective knock-down mutants, as described in the legend to **Fig. 1A**.

suggesting that depletion of intra-mycobacterial cAMP in WT-Rv0805 is indeed attributable to phosphodiesterase activity of Rv0805. The corresponding mRNA levels of PDEs (wild-type and the mutant) are over-expressed approximately 4.5-6-fold relative to the genomic *rv0805* level of WT-H37Rv (Figure 3-figure supplement 1A). In contrast, other PDE encoding genes (*rv1357* and *rv2387*), under identical conditions, demonstrate comparable expression levels in WT-H37Rv and *rv0805* over-expressing strains. Over-expression of these PDEs did not influence bacterial growth under normal conditions (Figure 3 - figure supplement 1B).

To further probe regulation of Rv0805 expression and its control of intra-mycobacterial cAMP level, we utilized a previously reported CRISPRi based approach (Singh *et al*, 2016) to construct *rv0805* and *phoP* knockdown (*rv0805*-KD, and *phoP-KD*, respectively) mutants. Consistent with *phoPR*-KO, *phoP*-KD shows a significantly higher *rv0805* expression in the presence of ATc relative to its absence (**Fig. 3D**). However, despite a significant down regulation of *rv0805* expression in presence of ATc, a comparable *phoP* expression was observed in *rv0805*-KD mutant both in the absence or presence of ATc. As a control, we observed a comparable expression of 16S rRNA in both knock-down mutants. Next, we determined intra-mycobacterial cAMP of the mutants as described in **Fig. 1** (**Fig. 3E**). cAMP level of *phoP*-KD (showing activation of Rv0805) was significantly lower relative to WT bacteria. In contrast, *rv0805*-KD mutant demonstrated a significantly higher level of cAMP relative to WT. We speculate that effective knocking down of *phoP* or *rv0805* is not truly reflected in the extent of variation of cAMP levels possibly due to the presence of numerous other mycobacterial PDEs. Together, these data represent an integrated view of our results suggesting that PhoP-dependant repression of *rv0805* regulates intra-mycobacterial cAMP level. In keeping with these results, activated PhoP under acidic pH conditions significantly represses *rv0805*, and intracellular mycobacteria most likely utilizes a higher level of cAMP to effectively mitigate stress for survival under stressful environment of the phagosome.

PhoP contributes to mycobacterial stress tolerance and intracellular survival by repressing the *rv0805* PDE expression

To investigate whether cAMP level influences mycobacterial susceptibility to stress, we compared *in vitro* growth under acidic pH (pH 4.5) (**Fig. 4A**). As expected, *phoPR*-KO showed a significant growth inhibition relative to WT under low pH (pH 4.5) (Bansal *et al*., 2017). WT- Rv0805, but not WT-Rv0805M displayed a comparable susceptibility to acidic pH as that of *phoPR*-KO. However, all four mycobacterial strains showed comparable growth at pH 7.0. Next, to compare growth of WT-Rv0805 and WT under oxidative stress, cells were grown in presence of increasing concentrations of diamide, a thiol-specific oxidant, and examined by microplate-based Alamar Blue assays (Figure 4 - figure supplement 1A). We have recently shown that *phoPR*-KO is significantly more sensitive to diamide relative to WT (Goar *et al*., 2022). Here, we uncovered that similar to *phoPR*-KO, WT-Rv0805, but not WT-Rv0805M was significantly more susceptible to diamide stress as compared to WT (**Fig. 4B**). A previous study had reported that *phoP*-deleted mutant strain was more sensitive to Cumene Hydrogen Peroxide (CHP), suggesting a role of PhoP in regulating mycobacterial stress response to oxidative stress (Walters *et al*., 2006). To compare sensitivity to CHP, we grew mycobacterial strains in presence of 50 µM CHP for 24 hours and determined their survival by enumerating CFU values (**Fig. 4C**). In this case we were unable to perform Alamar Blue-based survival assays requiring a longer time because of the bactericidal property of CHP. Our CFU data highlight that WT-Rv0805, but not WT-Rv0805M, displayed a significantly higher growth inhibition relative to WT in the presence of CHP. Together, these results reveal similar behaviour of *phoPR*-KO, and WT-Rv0805 by demonstrating a comparably higher susceptibility of these strains to acidic pH and oxidative stress relative to WT bacteria and indicate a link between intra-mycobacterial cAMP level and bacterial stress response. Collectively, these data suggest that at least one of the mechanisms by which PhoP contributes to global stress response is attributable to maintenance of cAMP level A previous study showed that *rv0805* over-expression in *M. smegmatis* influences cell wall permeability (Podobnik *et al*, 2009). Having shown a significantly higher sensitivity of WT-

Rv0805 to low pH and oxidative stress (relative to WT), we sought to investigate whether altered cell wall structure/properties of the mycobacterial strain contribute to elevated stress sensitivity. We compared expression level of lipid biosynthetic genes, which encode part of cell-wall structure of the bacilli (Gonzalo Asensio *et al.*, 2006a; Walters *et al.*, 2006). Our results suggest that in contrast to *phoPR*-KO, both WT-Rv0805 and WT-Rv0805M share a comparable expression profile of complex lipid biosynthesis genes as that of WT (Figure 4 - figure supplement 1B). Together, these results suggest that both strains expressing wild type or mutant PDEs share a largely similar cell-wall properties and are consistent with (a) a recent study reporting no significant effect of cAMP dysregulation on mycobacterial cell wall structure/permeability (Wong *et al*, 2023), and (b) role of PhoP in cell wall composition and complex lipid biosynthesis (Gonzalo Asensio *et al.*, 2006a; Goyal *et al.*, 2011; Walters *et al.*, 2006). Together, our findings facilitate an integrated view of our results, suggesting that higher susceptibility of WT-Rv0805 to stress conditions, is attributable to its reduced cAMP level.

To investigate the impact of mycobacterial cAMP level *in vivo*, we studied infection of murine macrophages using WT, WT-Rv0805, and WT-Rv0805M (**Fig. 4D**). In this assay, WT-H37Rv inhibits phagosome maturation, whereas phagosomes with *phoPR*-KO mature into phagolysosomes (Anil Kumar *et al.*, 2016). In our present experimental set up, although WT bacilli inhibited phagosome maturation, infection of macrophages with WT-Rv0805 and *phoPR*-KO matured into phagolysosomes, suggesting increased trafficking of the bacilli to lysosomes. Under identical conditions, WT-Rv0805M could effectively inhibit phagosome maturation just as WT bacteria. Results from co-localization experiments are plotted as **Fig. 4E** and as Pearson's correlation co-efficient of the quantified co-localization signals as **Fig. 4F**. These data suggest reduced ability of WT-Rv0805, but not WT-Rv0805M (relative to WT) to inhibit phagosome maturation. From these results, we suggest that ectopic expression of *rv0805* impacts phagosome maturation arguing in favour of a role of PhoP in influencing phagosome-lysosome fusion in macrophages.

Intra-bacterial cAMP level and its effect on *in vivo* survival of mycobacteria

To examine the effect *in vivo*, mice were infected with mycobacterial strains via the aerosol route. Day 1 post-infection, CFU analyses revealed a comparable count of four mycobacterial strains (~100 bacilli) in the mice lungs. However, for WT-Rv0805, the CFU recovered from infected lungs 4 weeks post-infection declined by ≍218-fold relative to the lungs infected with WT bacteria (**Fig. 5A**). In contrast, the CFU recovered from infected lungs after 4 weeks of infection by WT-Rv0805M marginally declined by ≍7-fold relative to the lungs infected with WT bacilli. These results suggest a significantly compromised ability of WT-Rv0805 (relative to WT) to replicate in the mice lungs. Note that *phoPR*-KO, under the conditions examined, showed a ≍ 246-fold lower lung burden compared to WT. In keeping with these results, while the WT bacilli disseminated to the spleens of infected mice, a significantly lower count of WT-Rv0805 was recovered from the spleens after 4 weeks of infection (**Fig. 5B**). Thus, we suggest that one of the reasons which accounts for an attenuated phenotype of *phoPR*-KO in both cellular and animal models is attributable to PhoP-dependent repression of *rv0805* PDE activity, which controls mycobacterial cAMP level.

M. tuberculosis H37Rv persists within granulomas where it is protected from the anti-mycobacterial immune effectors of the host. Histopathological evaluations showed that WT bacilli - infected lung sections displayed aggregation of granulocytes within alveolar spaces which degenerate progressively to necrotic cellular debris. In contrast, *phoPR*-KO and WT-Rv0805 showed less severe pathology as indicated by decreased tissue consolidation, smaller granulomas, and open alveolar space (**Fig. 5C**). Together, these results suggest that ectopic expression of

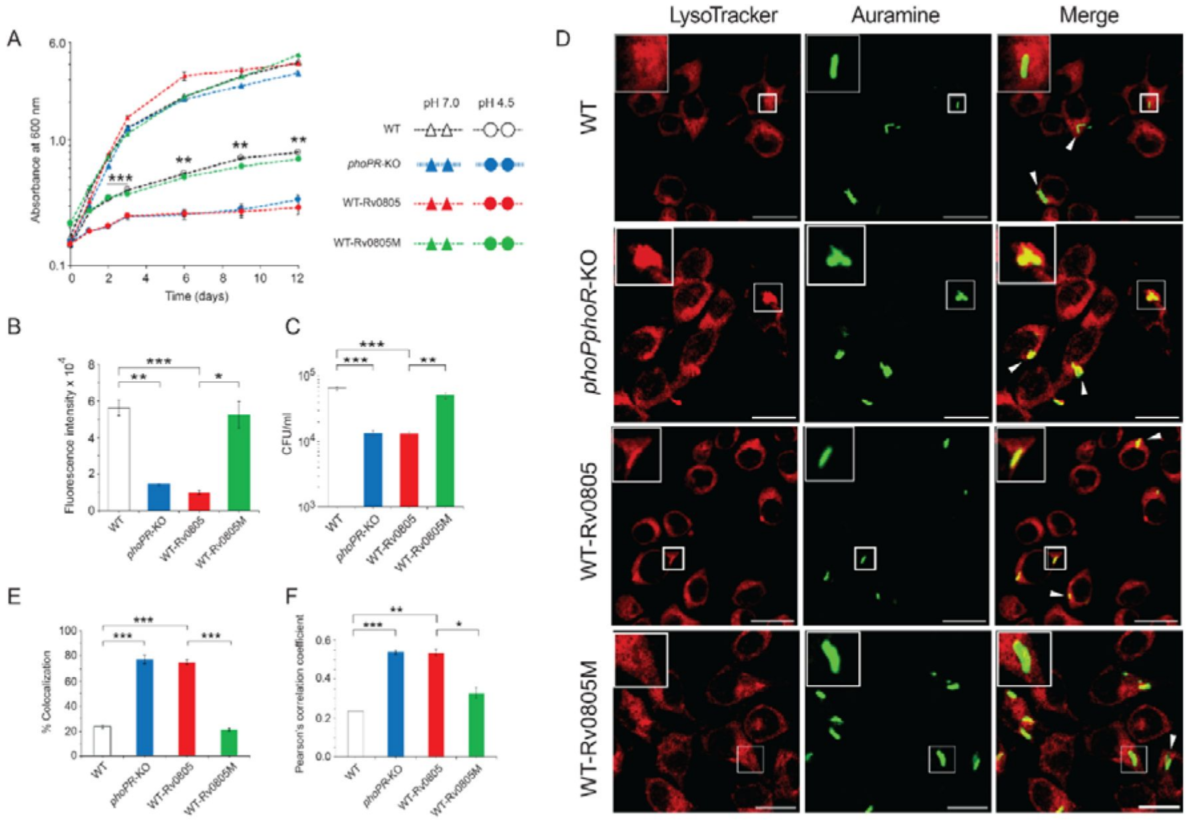

Figure 4

Regulation of cAMP level and its effect on mycobacterial stress tolerance and survival in macrophages.

(A) To compare susceptibility to low pH conditions, indicated mycobacterial strains were grown at pH 4.5, and similar to *phoPR*-KO (grey circles), WT-Rv0805 (red circles) shows a significant growth defect relative to WT (empty circles). However, WT-Rv0805M (green circles) grows comparably well as that of the WT (empty circles). In contrast, at pH 7.0 all four mycobacterial strains (WT, empty triangles; *phoPR*-KO, grey triangles; WT-Rv0805, red triangles; WT-Rv0805M, green triangles) displayed comparable growth. (B) Microplate-based assay using Alamar Blue was utilized to examine mycobacterial sensitivity to increasing concentrations of diamide. In this assay, reduction of Alamar Blue correlates with the change of a non-fluorescent blue to a fluorescent pink appearance, which is directly proportional to bacterial growth. Survival of indicated mycobacterial strains, under normal conditions and in presence of 5 mM diamide, were determined by plotting fluorescence intensity. The data is normalized relative to WT grown in presence of 5 mM diamide. (C) To compare susceptibility to stress conditions, these mycobacterial strains were next grown in the presence of 50 µM Cumene Hydrogen Peroxide (CHP). In presence of CHP, WT-Rv0805 (red column) but not WT-Rv0805M (green column), shows a significant growth defect [relative to WT (empty column)] in striking similarity to *phoPR*-KO (grey column). The growth experiments were performed in biological duplicates, each with two technical replicates (**$P \leq 0.01$; ***$P \leq 0.001$). (D) Murine macrophages were infected with indicated *M. tuberculosis* H37Rv strains. The cellular organelle was made visible by LysoTracker; mycobacterial strains were stained with phenolic auramine solution, and the confocal images display merge of two fluorescence signals (Lyso Tracker: red; H37Rv: green; scale bar: 10 µm). The insets in the merge panels indicate bacteria which either have inhibited or facilitated trafficking into lysosomes. White *arrowheads* in the merge panels indicate auramine labeled bacteria. (E) Bacterial co-localization of *M. tuberculosis* H37Rv strains. The percentage of auramine labelled strains co-localized with Lysotracker was determined by counting at least 100 infected cells in 10 different fields. The results show the average values with standard deviation determined from three independent experiments (***$P \leq 0.001$). (F) Pearson's correlation coefficient of images of internalized auramine-labelled mycobacteria and Lysotracker red marker in RAW 264.7 macrophages. Data are representative of mean ± S.D., derived from three independent experiments (*$P<0.05$; ***$P<0.001$).

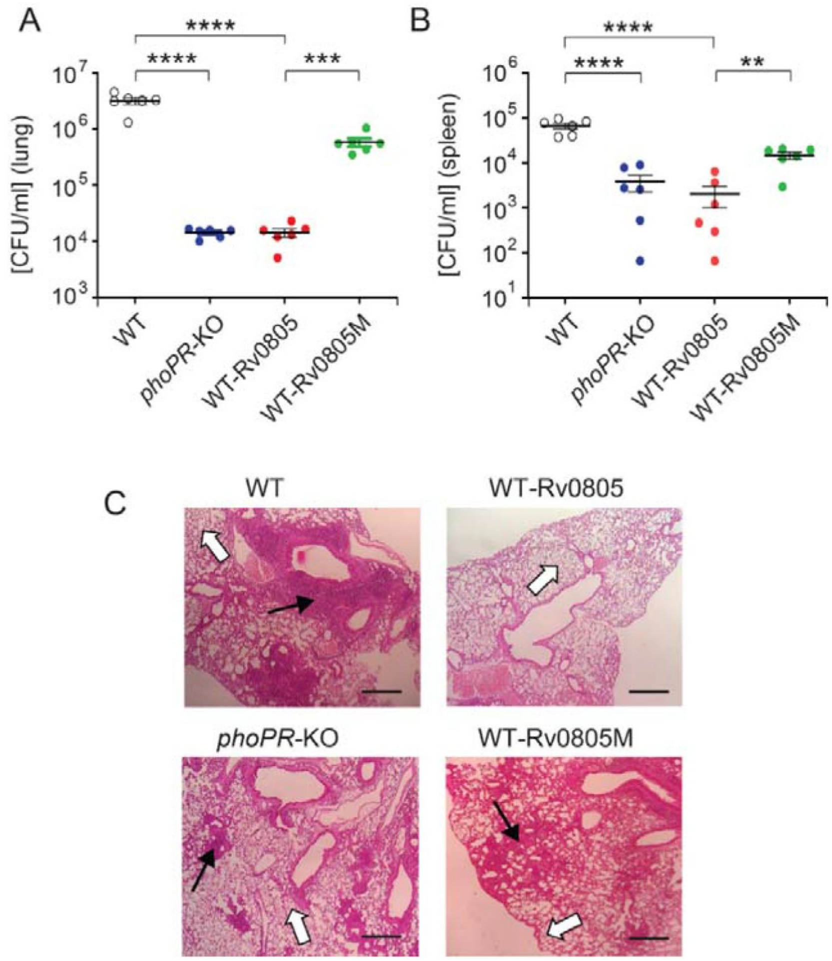

Figure 5

Dysregulation of mycobacterial cAMP level impacts mycobacterial survival *in vivo*.

(A-B) Survival of mycobacterial strains in mice (A) lung, and (B) spleen after animals were given an aerosol infection with ~100 CFU / lung. Mycobacterial load represents mean CFU values with standard deviations obtained from at least five animals per strains used (**P< 0.01; ***P<0.001; ****P<0.0001). (C) Histopathology of lung sections after 4 weeks of infection with indicated bacterial strains. Sections were stained with hematoxylin and eosin, observed under a light microscope, and images of the pathology sections collected at x40 magnification display granulomas (*filled arrows*) and alveolar space (*empty arrows*) (scale bar, 200 μm).

rv0805 in WT bacilli is somewhat functionally equivalent to deletion of *phoP*, suggesting that failure to maintain cAMP level most likely accounts for attenuated phenotype of the bacilli and absence of immunopathology in the lungs of infected mice.

Discussion

A number of studies suggest that conditions associated with host environment like low pH, and macrophage interactions often influence mycobacterial cAMP levels (Bai *et al*., 2009; Gazdik & McDonough, 2005). Although many bacterial pathogens modulate host cell cAMP levels as a common strategy, the mechanism of regulation of mycobacterial cAMP-level remains unknown. In this study, we sought to investigate whether PhoP, a master regulator implicated in controlling diverse mycobacterial stress response, is connected to mycobacterial cAMP level. We find that under normal conditions as well as under carefully controlled single stress conditions, *phoPR*-KO shows a significantly lower level of cAMP relative to the WT bacilli (**Fig. 1A**), and complementation of the mutant restored cAMP level. To investigate the mechanism, we next probed regulation of adenylate cyclases and PDEs (**Fig. 2**) and demonstrated that PhoP functions as a major repressor of *rv0805*, encoding cAMP specific PDE. Indeed, this newly-identified *rv0805* regulation, coupled with a very recent discovery that phosphodiesterase activity of Rv0805 controls propionate detoxification (McDowell *et al*, 2023), fits well with and explains, the previously puzzling *in vivo* observation by Abramovitch et al (Abramovitch *et al*., 2011) that PhoP-controlled *aprABC* locus is associated with the regulation of genes of carbon and propionate metabolism.

Although a large number of ACs are present in *M. tuberculosis* genome, a class III metallo-phosphoesterase Rv0805 was earlier considered the only PDE, specific for mycobacterial cAMP. However, a recent study has identified an atypical class II PDE Rv1339, which upon overexpression reduces cAMP level and contributes to antibiotic sensitivity (Thomson *et al*., 2022). While, the functional role of Rv1339 in *M. tuberculosis* is yet to be understood, crystal structure and biochemical evidence suggest that dimeric Rv0805 is stabilized by the presence of a divalent cation, and remains catalytically active on a broad range of linear and cyclic PDE substrates *in vitro* (Keppetipola & Shuman, 2008; Shenoy *et al*., 2007; Shenoy *et al*., 2005). More recently, cyclic nucleotide hydrolytic activity of mycobacterial Rv0805 has been implicated in propionate detoxification (McDowell *et al*., 2023). However, the mechanisms of regulation of Rv0805 and maintenance of mycobacterial cAMP level remain unknown.

To examine biological significance of PhoP-dependent Rv0805 expression, we studied *rv0805* expression under acidic conditions of growth as *phoPR* system is induced under acidic pH both *in vitro* and in macrophages (Abramovitch *et al*., 2011; Bansal *et al*., 2017). A significantly higher repression of *rv0805* expression under acidic pH relative to normal conditions is consistent with activation of PhoP and subsequent repression of *rv0805* (**Fig. 3**). These results further suggest that effective mitigation of stress by mycobacteria possibly requires a higher cAMP level for survival under intra-phagosomal environment. In keeping with these results, we find that (a) P~PhoP binds to *rv0805* regulatory region (Fig. 2-figure supplement 1), and (b) PhoP-dependent *rv0805* expression requires PhoR (**Figs. 3A-B**), the cognate kinase which activates PhoP in a signal-dependent manner (Gupta *et al*., 2006; Singh *et al*, 2023). These results account for a consistently lower level of cAMP in *phoPR*-KO relative to the WT bacilli. Notably, except recently reported PDE Rv1339, Rv0805 has been known as the only cAMP-specific PDE present in the slow growing pathogenic mycobacteria and its closely related species (Matange, 2015; Shenoy *et al*., 2007; Shenoy *et al*., 2005), and Rv1339 expression does not appear to be regulated by the *phoP* locus (**Fig. 2**). Thus, the above results showing PhoP-dependent repression of *rv0805* activity likely represents the most critical step of regulation of mycobacterial cAMP level. In this connection, our recent results that PhoP interacts with cAMP receptor protein, CRP and a complex of two interacting regulators control expression of virulence determinants (Khan *et al*, 2022), invite speculation of a complex regulatory control of cAMP-responsive mycobacterial physiology.

As one might argue that PhoP deletion and *rv0805* over-expression could be unrelated and independent events, we constructed *phoP* and *rv0805* knock-down mutants to further investigate the PhoP-Rv0805-cAMP pathway. Our objective was to probe regulation of expression (**Fig. 3D**) and examine the impact on mycobacterial cAMP (**Fig. 3E**). *phoP*-KD significantly elevated *rv0805* expression; however, *phoP* expression remains unaffected in *rv0805*-KD (**Fig. 3D**). While elevated *rv0805* expression in *phoP*-KD reduces cAMP level, understandably cAMP level is elevated in *rv0805*-KD mutant (**Fig. 3E**). These results integrate PhoP-dependent *rv0805* repression with mycobacterial cAMP level, suggesting how *phoP*-deletion or knock-down, at least in part, mimics Rv0805 over-expression. These considerations take on more significance given the fact that these two events have similar consequences on relevant strains with respect to stress tolerance, and survival in cellular and animal models. Thus, our results suggest that ectopic expression of *rv0805* is functionally equivalent to deletion of the *phoP* locus. This observation is in apparent conflict with a previous work by Matange and collaborators (Matange *et al*., 2013), suggesting a cAMP-independent transcriptional response in *rv0805* over-expressing *M. tuberculosis* H37Rv. Although both studies were performed with *rv0805* over-expressing bacilli, the fact that important differences in the expression of PDEs in this study (Matange *et al*., 2013) and in our assays – yielding significantly different levels of *rv0805* expression - most likely account for this discrepancy. While we cannot completely rule out the possibility of cleavage of other cyclic nucleotide(s) by Rv0805 (Keppetipola & Shuman, 2008; Shenoy *et al*., 2007; Shenoy *et al*., 2005) impacting our results, consistent with a previous study our results correlate *rv0805* expression with intra-mycobacterial cAMP level (Agarwal *et al*., 2009). Further, our data on the effect of expression of cyclic nucleotide-specific PDE Rv0805 or its inactive mutant (Rv0805M) correlate well with enzyme activities of the corresponding PDEs on mycobacterial cAMP levels (**Fig. 3C**). Thus, we infer that PhoP-dependent regulation of *rv0805* is a critical regulator of intra-mycobacterial cAMP level.

Our experiments to understand physiological significance of PhoP-dependent repression of *rv0805* expression uncovers a comparable stress tolerance of WT-Rv0805 and *phoPR*-KO (significantly reduced relative to WT) (**Fig. 4**). These results are consistent with the notion that cAMP level, at least in part, accounts for mycobacterial stress response. Along the line, WT-Rv0805 displayed a reduced ability to inhibit phagosome-lysosome fusion like *phoPR*-KO (**Fig. 4**). Further, we show that WT-Rv0805, unlike the WT bacilli or WT-Rv0805M, shows a significantly reduced intracellular growth in mice as that of *phoPR*-KO (**Fig. 5**). Thus, these results are of fundamental significance to establish that PhoP contributes to maintenance of cAMP level and integrates it to mechanisms of mycobacterial stress tolerance and intracellular survival. Together, we identify a novel mycobacterial pathway as a therapeutic target and provide yet another example of an intimate link between bacterial physiology and intracellular survival of the tubercle bacilli.

The results reported here are presented schematically in **Fig. 6**. In summary, upon sensing low acidic pH as a signal PhoR activates PhoP, P~PhoP binds to *rv0805* upstream regulatory region and functions as a specific repressor of Rv0805. Therefore, we observed (a) a reproducibly lower level of cAMP in *phoPR*-KO relative to WT-H37Rv, (b) a significantly reduced expression of *rv0805* in WT-H37Rv, grown under acidic pH relative to normal conditions, and (c) comparable cAMP levels in *phoPR*-KO and WT-Rv0805. This is why the two strains remain ineffective to mount an appropriate stress response, most likely due to their inability to coordinate regulation of gene expression because of dysregulation of intra-mycobacterial cAMP level. However, without uncoupling regulatory control of PhoPR and *rv0805* expression, we cannot confirm that dysregulation of cAMP level accounts for virulence attenuation of *phoPR*-KO. Given the fact that *rv0805*-depleted *M. tuberculosis* is growth attenuated *in vivo* (McDowell et al., 2023), paradoxically ectopic expression of *rv0805* leads to dysregulated metabolic adaptation, thereby resulting in reduced stress tolerance and intracellular survival.

Materials and Methods

Bacterial strains and culture conditions

E. coli DH5α was used for cloning experiments. *E. coli* BL21(DE3), an *E. coli* B strain lysogenized with λDE3, a prophage expressing T7 RNA polymerase from the IPTG (isopropyl-β-D-thiogalactopyranoside)-inducible *lac*UV5 promoter (Studier & Moffatt, 1986) was used as the host for over-expression of recombinant proteins. *E. coli* strains were grown in LB medium at 37°C with shaking, transformed according to standard procedures and the transformants were selected on media containing appropriate antibiotics plates. WT- and mutant *M. tuberculosis* and *M. smegmatis* mc^2155 were grown at 37°C in Middlebrook 7H9 broth (Difco) containing 0.2% glycerol, 0.05% Tween-80 and 10% ADC (albumin-dextrose-catalase) or on 7H10-agar medium (Difco) containing 0.5% glycerol and 10% OADC (oleic acid-albumin-dextrose-catalase) enrichment. *phoPR* disruption mutant of *M. tuberculosis* H37Rv (*phoPR*-KO, a kind gift of Dr. Issar Smith) was constructed as described (Walters *et al*., 2006). To this end, a kanamycin-resistant cassette from pUC-K4 was inserted into a unique EcoRV site within the coding region of *phoP* gene, and disruption was confirmed by Southern blot analysis of chromosomal DNA isolated from the mutant. Next, purified plasmid DNAs were electroporated into wild-type *M. tuberculosis* strain by standard protocol (Jacobs *et al*, 1991). To complement *phoPR* expression, pSM607 containing a 3.6-kb DNA fragment of *M. tuberculosis phoPR* including 200-bp *phoP* promoter region, a hygromycin resistance cassette, *attP* site and the gene encoding phage L5 integrase, as detailed earlier (Walters *et al*., 2006) was used to transform *phoPR* mutant to integrate at the L5 *attB* site. Growth, transformation of wild-type (WT), *phoPR*-KO, the complemented mutant (Compl.) *M. tuberculosis* and selection of transformants on appropriate antibiotics plates were performed as described (Goyal *et al*., 2011). When appropriate, antibiotics were used at the following concentrations: hygromycin (hyg), 250 μg/ml for *E. coli* or 50 μg/ml for mycobacterial strains; streptomycin (str), 100 μg/ml for *E. coli* or 20 μg/ml for mycobacterial strains; kanamycin (kan), 20 μg/ml for mycobacterial strains. For *in vitro* growth under specific stress conditions, indicated mycobacterial strains were grown to mid log phase (OD_{600} 0.4-0.6) and exposed to different stress conditions. For acid stress, cells were initially grown in 7H9 media, pH7.0 and on attaining mid log phase it was transferred to acidic media (7H9 media, pH 4.5), and grown for further two hours at 37°C. For oxidative stress, cells were grown in presence of 50 μM CHP (Sigma) for 24 hours or indicated diamide concentration(s) for 7 days. For NO stress, cells grown to mid log phase were exposed to 0.5 mM DataNonoate for 40 minutes (Voskuil *et al*, 2003).

cAMP measurement

Mycobacterial cell pellets were collected and washed with 1x PBS buffer, cells were resuspended in IP buffer (50 mM Tris pH 7.5, 150 mM NaCl, 1 mM EDTA pH 8.0, 1 mM PMSF, 5% glycerol and 1% TritonX 100) and cell lysates (CL) were prepared by lysing the cells in presence of Lysing Matrix B (100 μm silica beads; MP Bio) using a FastPrep-24 bead beater (MP Bio) at a speed setting of 6.0 for 30 seconds. The procedure was repeated for 10 cycles with incubation on ice in between pulses. The supernatant was collected by centrifugation at 13,000 rpm for 10 minutes and filtered through 0.22 μm filter (Millipore). cAMP levels in the cells were determined in a plate reader by using fluorescence-based cAMP detection kit (Abcam) according to the manufacturer's recommendations and normalized to the total protein present in the samples as determined by a BCA protein estimation kit (Pierce). For secretion studies, each mycobacterial strain was grown in Sauton's media as described (Anil Kumar *et al*., 2016), comparable counts of bacterial cells were

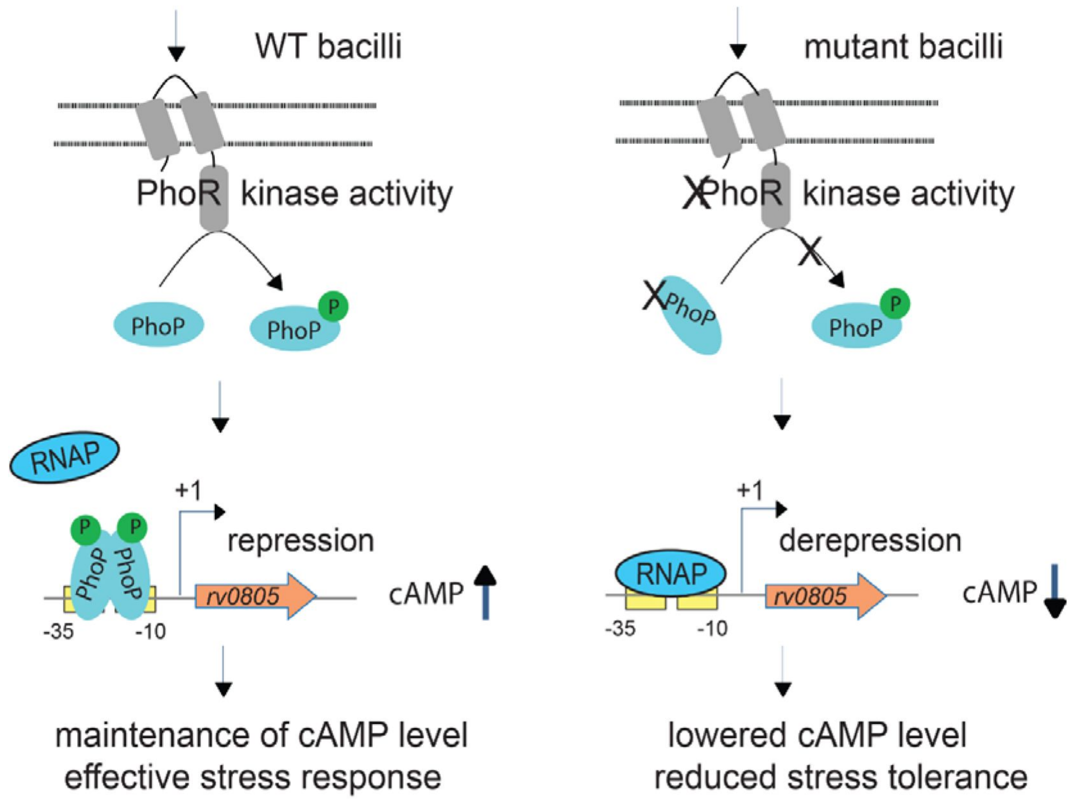

Figure 6

Increased cAMP level and effective stress response versus decreased cAMP level and reduced stress tolerance of mycobacteria.

In this model, upon activation by an appropriate signal via the cognate sensor PhoR, P∼PhoP binds to *rv0805* regulatory region and functions as a specific repressor by preventing access for mycobacterial RNA polymerase (RNAP) to bind to the promoter and initiate transcription. In keeping with PhoP-dependent *rv0805* repression, our results demonstrate a reproducibly lower level of cAMP in *phoPR*-KO relative to WT bacilli. Thus, *phoPR*-KO (or WT-Rv0805) remains ineffective to mount an appropriate stress response most likely due to its inability to coordinate regulated gene expression because of dysregulation of cAMP level, accounts for their reduced stress tolerance. Together, these molecular events suggest that failure to maintain cAMP **level** accounts for attenuated phenotype of the bacilli.

pelleted, resuspended in 2 ml of Sauton's media in a 6-well plate format for 2 hours at 37°C, and the supernatants (culture filtrates, CF) were collected for cAMP measurements, as described previously (Anil Kumar *et al*., 2016).

Cloning

M. tuberculosis full-length ORFs of interest were cloned between EcoRI and HindIII sites of the mycobacterial expression vector pSTKi (Parikh *et al*., 2013) and expressed from the P_{myc1}*tetO* promoter. Mutation in Rv0805 was introduced by two-stage overlap extension method using mutagenic primers (Supplementary file 1b), and the construct was verified by DNA sequencing. For over-expression of WT-or mutant PDEs, WT bacilli was transformed with pST-rv0805 or pST-rv0805M to generate WT-Rv0805 or WT-Rv0805M, respectively.

Emsa

rv0805up DNA probe was PCR amplified, resolved on an agarose gel, recovered by gel extraction, and end-labelled with [γ-^{32}P ATP] (1000 Ci nmol^{-1}) using T4 polynucleotide kinase. The end-labelled DNA probe was purified from free label by Sephadex G-50 spin columns (GE Healthcare), and incubated with increasing amounts of purified PhoP in a total volume of 10 µl binding mix (50 mM Tris-HCl, pH 7.5, 50 mM NaCl, 0.2 mg/ml of bovine serum albumin, 10% glycerol, 1 mM dithiothreitol, ≍ 50 ng of labelled DNA probe, and 0.2 µg of sheared herring sperm DNA) at 20°C for 20 minutes. DNA-protein complexes were resolved by electrophoresis on a 6% (w/v) polyacrylamide gel (non-denaturing) in 0.5X TBE (89 mM Tris-base, 89 mM boric acid and 2 mM EDTA) at 70 V and 4°C, and the position of the radioactive material was determined by exposure to a phosphor storage screen.

Construction of *M. tuberculosis phoP* and *rv0805* knock-down mutants

In this study, we utilized a previously reported CRISPRi system (Singh *et al*., 2016) to construct knock-down mutants of *phoP* and r*v0805* (*phoP*-KD, and *rv0805*-KD, respectively). This approach efficiently inhibits expression of target genes via inducible expression of dCas9 along with gene specific guide RNAs (sgRNA). The RNAs were 20 nt long and complementary to the non-template strand of the target gene. The sgRNAs of *phoP* and *rv0805* were cloned in a vector pRH2521 using BbsI enzyme and the constructs were confirmed by sequencing. The corresponding clones were used to transform *M. tuberculosis* harbouring pRH2502, a vector expressing an inactive version of *Streptococcus pyogenes* cas9 (dcas9).

To express dcas9 and repress sgRNA-targeted genes (*phoP* or *rv0805*), the bacterial strains were grown with or without 600 ng/ml of anhydro-tetracycline (ATc) every 48 hours, and cultures were grown for 4 days. RNA isolation was carried out, and RT-qPCR experiments verified significant repression of target genes. For the induced strains (in presence of ATc) expressing sgRNAs targeting +155 to +175 (relative to *phoP* translational start site) and +224 to +244 sequences (relative to *rv0805* translational start site), we obtained approximately 85% and 90% reduction of *phoP* and *rv0805* RNA abundance, respectively, relative to corresponding un-induced strains. The oligonucleotides used to generate gene-specific sgRNA constructs and the plasmids utilized in knock-down experiments are listed in Supplementary file 1b.

RNA isolation

Total RNA was extracted from exponentially growing bacterial cultures grown with or without specific stress as described above. Briefly, 25 ml of bacterial culture was grown to mid-log phase (OD_{600}= 0.4 to 0.6) and combined with 40 ml of 5 M guanidinium thiocyanate solution containing 1% β-mercaptoethanol and 0.5% Tween 80. Cells were pelleted by centrifugation, and lysed by re-suspending in 1 ml Trizol (Ambion) in the presence of Lysing Matrix B (100 µm silica beads; MP

Bio) using a FastPrep-24 bead beater (MP Bio) at a speed setting of 6.0 for 30 seconds. The procedure was repeated for 2-3 cycles with incubation on ice in between pulses. Next, cell lysates were centrifuged at 13000 rpm for 10 minutes; supernatant was collected and processed for RNA isolation using Direct-Zol^{TM} RNA isolation kit (ZYMO). Following extraction, RNA was treated with DNAse I (Promega) to degrade contaminating DNA, and integrity was assessed using a Nanodrop (ND-1000, Spectrophotometer). RNA samples were further checked for intactness of 23S and 16S rRNA using formaldehyde-agarose gel electrophoresis, and Qubit fluorometer (Invitrogen).

Quantitative Real-Time PCR

cDNA synthesis and PCR reactions were carried out using total RNA extracted from each bacterial culture, and Superscript III platinum-SYBR green one-step qRT-PCR kit (Invitrogen) with appropriate primer pairs (2 μM) using an ABI real-time PCR detection system. Oligonucleotide primer sequences used in RT-qPCR experiments are listed in Supplementary file 1a. Control reactions with platinum Taq DNA polymerase (Invitrogen) confirmed absence of genomic DNA in all our RNA preparations, and endogenously expressed *M. tuberculosis rpoB* was used as an internal control. Fold difference in gene expression was calculated using $\Delta\Delta C_T$ method (Schmittgen & Livak, 2008). Average fold differences in mRNA levels were determined from at least two biological repeats each with two technical repeats. Non-significant difference is not indicated.

ChIP assays

ChIP experiments in actively growing cultures of *M. tuberculosis* were carried out essentially as described previously (Fol *et al*, 2006). Immunoprecipitation (IP) was performed using anti-PhoP antibody and protein A/G agarose beads (Pierce). qPCR reactions included PAGE purified primer pairs (Supplementary file 1a) spanning specific promoter regions using suitable dilutions of immunoprecipitated (IP) DNA in a reaction buffer containing SYBR green mix, and one unit of Platinum Taq DNA polymerase (Invitrogen). An IP experiment without adding antibody (mock) was used as the negative control, and data was analysed relative to PCR signal from the mock sample. PCR amplifications were carried out for 40 cycles using serially diluted DNA samples (mock, IP treated and total input) in a real-time PCR detection system (Applied Biosystems). In all cases melting curve analysis confirmed amplification of a single product.

Immunoblotting

Cell lysates or culture filtrates were resolved by 12% SDS-PAGE and visualized by Western blot analysis using appropriate antibodies. Briefly, resolved samples were electroblotted onto polyvinyl difluoride (PVDF) membranes (Millipore, USA) and were detected by anti-GroEL2 antibody (Sigma), anti CFP-10 antibody (Abcam) or affinity-purified anti-PhoP antibody elicited in rabbit (Alpha Omega Sciences, India). Goat anti-rabbit secondary antibody conjugated to horseradish peroxidase was used, and blots were developed with Luminata Forte Chemiluminescence reagent (Millipore, USA). RNA polymerase was used as a loading control and was detected with monoclonal antibody against β-subunit of RNA polymerase, RpoB (Abcam).

Alamar Blue assay

In this assay, reduction of Alamar Blue correlates with the change of a non-fluorescent blue to a fluorescent pink appearance, which is directly linked to bacterial growth. *M. tuberculosis* H37Rv was grown in 7H9 media (Difco) with 10% ADS (albumin, dextrose and NaCl) to an OD_{600} of 0.4, and freshly diluted to OD_{600} of 0.02. Next, increasing concentrations of diamide was added to the wells of a 96-well plate containing 0.05 ml 7H9 media followed by addition of 0.05 ml of *M. tuberculosis* H37Rv culture (0.02 OD_{600}). The plate was incubated at 37°C for 7 days. Finally, 0.02 ml of 0.02% Resazurin (sodium salt, MP Bio), prepared in sterile 7H9 media was added to each of the wells and the change in colour was examined after incubation at 37°C for 16 hours. The fluorescence excitation was at 530 nm and emission was recorded at 590 nm. Efficiency of inhibition was calculated relative to control wells which did not include diamide, and Rifampicin was included as a positive control to confirm validity of the assay.

Macrophage Infections

Virulence of indicated H37Rv strains were assessed in murine macrophages according to the previously published protocol (Solans *et al*., 2014). Briefly, RAW264.7 macrophages were grown in DMEM media containing 10% Fetal bovine serum at 37°C under 5% CO_2, and seeded onto #1 thickness, 18 mm diameter glass coverslips in a 12-well plate at a 40% confluency (0.5 million cells). Cells were independently infected with titrated cultures of WT, WT-Rv0805, WT-Rv0805M and *phoPR*-KO strains at a multiplicity of infection (MOI) of 1:5 for 3 hours at 37°C in 5% CO_2, followed by 1X PBS washes thrice. The macrophages were further incubated for 3 hours at 37°C. After infection, extracellular bacteria were removed by washing thrice with PBS. To visualize trafficking of the tubercle bacilli, mycobacterial strains were stained with phenolic auramine solution (which selectively binds to mycolic acids) for 15 minutes followed by washing with acid alcohol solution and finally with 1X PBS. The cells were stained with 150 nM LysoTracker Red DND-99 (Invitrogen) for 30 minutes in a CO_2 incubator. Next, the cells were fixed with 4% paraformaldehyde for 15 minutes, washed thrice with PBS, the coverslips were mounted in Slow Fade-Anti-Fade (Invitrogen) and analysed using laser scanning confocal microscope (Nikon) equipped with Argon (488 nm excitation line; 510 nm emission detection) and LD (561 nm excitation line; 594 nm emission detection) laser lines. Digital images were processed with IMARIS imaging software (version 9.20). Details of the experimental methods and the laser/detector settings were optimized using macrophage cells infected with WT-H37Rv as described previously (Anil Kumar *et al*., 2016). A standard set of intensity threshold was made applicable for all images, and percent bacterial co-localization was determined by analyses of at least 50 infected cells originating from 10 different fields of each of the three independent biological repeats.

Mouse infections

All experiments pertaining to mice were in accordance with Institutional regulations after review of protocols and approval by the Institutional Animal Ethics Committee (IAEC/17/05, and IAEC/19/02). Mice were maintained and bred in the animal house facility of CSIR-IMTECH. Animal infection studies and subsequent experiments were carried out in the Institutional BSL-3 facility as per institutional biosafety guidelines. Briefly, the experiments were conducted with 8-10 weeks old C57BL/6 mice, infected intranasally and euthanized post-infection for evaluation of bacterial load in lungs and spleens. Infections were given through the respiratory route using an inhalation exposure system (Glass-col) with passaged *M. tuberculosis* H37Rv cultures of mid log phase. The actual bacterial load delivered to the animals was enumerated from 3-5 aerogenically challenged mice, 1day post aerosol challenge. The animals were found to achieve a bacillary deposition of 100 to 200 CFU in the lungs for each strain. Four weeks post infection, the animals were sacrificed by cervical dislocation, lungs and spleens were isolated aseptically from the euthanized animals, homogenized in sterile 1X PBS and plated after serially diluting the lysates on 7H11 agar plates, supplemented with 10% OADC and antibiotics (50 µg/ml carbenicillin, 30 µg/ml polymyxin B, 10 µg/ml vancomycin, 20 µg/ml Trimethoprim, 20 µg/ml cycloheximide, and 20 µg/ml Amphotericin B) to enumerate CFU. For histopathology, left lung lobes were fixed in 10% buffered formalin, embedded in paraffin and stained with haematoxylin and eosin for visualization under the microscope. The level of pathology was scored by analysing perivascular cuffing, leukocyte infiltration, multinucleated giant cell formation and epithelial cell injury.

Statistical analysis

Data are presented as arithmetic means of the results obtained from multiple replicate experiments ± standard deviations. Statistical significance was determined by Student's paired t-test using Microsoft Excel or Graph Pad Prism. Statistical significance was considered at P values of 0.05.

Acknowledgements

We are grateful to G. Marcela Rodriguez and Issar Smith (PHRI, New Jersey Medical School - UMDNJ) for Δ*phoP*-H37Rv, and the complemented *M. tuberculosis* H37Rv strains, Adrie Steyn (University of Alabama) for pUAB300/pUAB400 plasmids, Ashwani Kumar for very helpful discussions, and Sanjeev Khosla for critical reading of the manuscript. We thank members of the Institutional animal facility (iCARE) for their help with approval of our projects from the Institutional Animal Ethics Committee. This study received financial support from intramural grants of CSIR-IMTECH (OLP-0170), CSIR (MLP-0049) and by a research grant (to D.S) from SERB (EMR/2016/004904), Department of Science and Technology (DST). H.K., P.P., H.G., B.B. and N.B. were supported by CSIR pre-doctoral fellowships.

Supplemental Data

The supplemental data include three (3) supplemental figures, and two supplemental tables (Supplemental files 1a, and 1b, respectively). All data generated or analysed during this study are included in the manuscript and supporting file; source data files have been provided for all relevant figures.

Competing interests

The authors declare that no competing interests exist.

Author contributions

H.K., P.P., H. G., B.B. and D.S. designed research; H.K., P. P., H.G., and B.B. performed research; N.B. contributed analytical tools; H.K., P. P., H. G., B.B., N.B., and D.S. analysed data; and D.S. wrote the manuscript.

Supplementary figures

Figure 1-figure supplement 1: Defined and indicated stress conditions do not influence *in vitro* growth of mycobacterial strains. To determine the impact of *in vitro* stress on bacterial growth under conditions when intra-mycobacterial cAMP levels were measured, WT bacilli, *phoPR*-KO, and the complemented mutant (Compl.) were exposed to carefully controlled stress conditions as detailed in the Methods, and CFUs enumerated. The results clearly suggest that bacterial viability under indicated conditions of stress do not impact intra-mycobacterial cAMP level.

Figure 2-figure supplement 1: Probing *in vitro* DNA binding of PhoP to the *rv0805* regulatory region. (A) Core binding site of PhoP within the promoter region spanning −150 to +1-bp upstream regulatory region of *rv0805* (rv0805up; relative to ORF start site) was identified by the MEME Suit bioinformatic software using previously-identified consensus PhoP binding sequence (**He and Wang, 2014**). Note that the predicted binding site spans from −110 to −127 relative to the ORF start site (p=0.000726). (B) To verify DNA binding *in vitro*, rv0805up was amplified, end-labelled, and EMSA was carried out using radio-labelled rv0805up for binding of increasing concentrations of PhoP (lanes 2-4) and PhoP, pre-incubated in phosphorylation mix with acetyl-phosphate (AcP) (lanes 5-7), respectively. Lane 1 shows the free probe. The binding mixtures in lanes 2-4, and lanes 5-7 contained indicated proteins at 1, 2 and 4 μM, respectively. The position of the radioactive

material was determined by exposure to a phosphor storage screen. *Open* and *filled* arrows indicate free probe and slower-moving complexes with band shifts produced in presence of P~PhoP, respectively.

Figure 3-figure supplement 1: Ectopic expression of wild-type and mutant *rv0805* (Rv0805M) in WT bacilli. (A) Real-time RT-qPCR was carried out to compare expression of indicated phosphodiesterases (PDEs) in WT (empty bar), WT-Rv0805 (red bar) and WT-Rv0805M (green bar), respectively, as described in the Methods. Average fold difference in mRNA levels from two biological repeats (each with a technical repeat) were determined as described in the Methods (**P≤0.01; ***P≤0.001). Nonsignificant difference is not indicated. (B) To compare growth of WT and the variant mycobacterial strains, CFU values were determined under normal conditions of growth as described in the methods.

Figure 4-figure supplement 1: PhoP-dependent *rv0805* expression contributes to mycobacterial survival under oxidative stress. (A) In this experiment, we compared metabolic activity of WT, *phoPR*-KO, WT-Rv0805, and WT-Rv0805M, grown in presence of increasing concentrations of diamide by using Alamar Blue assay. Because reduction of Alamar Blue correlates with the change of a non-fluorescent blue to a fluorescent pink appearance, mycobacterial metabolic activity could be assessed by monitoring fluorescence. Two following controls were included in the Alamar Blue assays: Rifampicin control, reflecting mycobacterial growth inhibition to confirm validity of the assay, and no bacteria control indicated at the left top corner of the plate. (B) Next, effect of over expression of *rv0805* on mycobacterial cell-wall structure was assessed by analysing expression of lipid biosynthetic genes of indicated mycobacterial strains by RT-qPCR. Average fold difference in mRNA levels from two biological repeats (each with a technical repeat) were determined as described in the Methods (***P≤0.001). Nonsignificant difference is not indicated.

Supplementary Tables

Supplementary file 1a

Sequences of oligonucleotide primers used in RT-qPCR and ChIP-qPCR measurements reported in this study

Supplementary file 1b

Sequences of oligonucleotide primers used for amplification and cloning, and plasmids used in this study

Source data files for supplementary figures

Figure 1-figure supplement 1

Figure 2-figure supplement 1

Figure 3-figure supplement 1

Figure 4-figure supplement 1

Supplemental Data

The supplemental data include four supplemental figures and two supplemental tables (supplementary files 1a and 1b, respectively). All other data are part of the paper and its supplemental files.

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Reviewer #1 (Public Review):

Summary:

This paper provides a straightforward mechanism of how mycobacterial cAMP level is increased under stressful conditions and shows that the increase is important for the survival of the bacterium in animal hosts. The cAMP level is increased by decreasing the expression of an enzyme that degrades cAMP.

Strengths:

The paper shows that under different stresses the response regulator PhoP represses a phosphodiesterase (PDE) that degrades cAMP specifically. Identification of PhoP as a regulator of cAMP is significant progress in understanding Mtb pathogenesis, as increase in cAMP apparently increases bacterial survival upon infection. On the practical side, reduction of cAMP by increasing PDE can be a means to attenuate the growth of the bacilli. The results have wider implications since PhoP is implicated in controlling diverse mycobacterial stress responses and many bacterial pathogens modulate host cell cAMP level. The results here are straightforward, internally consistent, and of both theoretical and applied interests. The results also open considerable future work, especially how increases in cAMP level help to increase survival of the pathogen.

Weaknesses:

It is not clear whether PhoP-PDE Rv0805 is the only pathway to regulate cAMP level under stress.

https://doi.org/10.7554/eLife.92136.2.sa1

Reviewer #2 (Public Review):

Summary: In the manuscript, the authors have presented new mechanistic details to show how intracellular cAMP levels are maintained linked to the phosphodiesterase enzyme which in turn is controlled by PhoP. Later, they showed the physiological relevance linked to altered cAMP concentrations.

Strengths: Well thought out experiments. The authors carefully planned the experiments well to uncover the molecular aspects of it diligently.

Weaknesses: Some fresh queries were made based on the author's previous responses and hope to get satisfactory answers this time.

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Author Response

The following is the authors' response to the original reviews.

> ***Public Review:***
>
> *Summary:*
>
> *This paper reports how mycobacterial cAMP level is increased under stressful conditions and that the increase is important in the survival of the bacterium in animal hosts.*
>
> *Strengths:*
>
> *The authors show that under different stresses the response regulator PhoP represses a phosphodiesterase (PDE) that degrades cAMP specifically. Identification of a PDE specific to cAMP is significant progress in understanding Mtb pathogenesis. An increase in cAMP apparently increases bacterial survival upon infection. On the practical side, the reduction of cAMP by increasing PDE can be a means to attenuate the growth of the bacilli. The results have wider implications since PhoP is implicated in controlling diverse mycobacterial stress responses and many bacterial pathogens modulate host cell cAMP level. The results here are straightforward, internally consistent, and of both theoretical and applied interests.*

We thank the reviewers for these extremely encouraging comments.

> *Weaknesses:*
>
> *Repression of PDE promoter by binding of phosphorylated PhoP could have been shown at higher precision. The binding is now somewhere along a roughly 500 bp region. Although the regulation of PDE is shown to be by transcriptional repression only, it has been described as a homeostatic mechanism. The latter would have required a demonstration of both repression and activation by negative feedback.*

We agree. We have now performed EMSA (Electrophoretic Mobility Shift Assay) experiments and included the data showing DNA binding of PhoP to the upstream regulatory region of rv0805 (rv0805up) as a supplemental figure (see Figure 2-figure supplement 1). The supplemental figure, figure caption, and the relevant results have been adjusted accordingly in the revised manuscript.

Further, as recommended by the reviewer we have now removed the term 'homeostatic mechanism' and rephrased it with 'maintenance of cAMP level' in the manuscript.

Response to Reviewers' comments

> ***Reviewer #1:***
>
> *The authors have used homeostasis inappropriately. Homeostasis usually requires negative feedback (a clear example is the regulation of Lambda prm promoter). Here, there is no feedback from changes in PDE or cAMP level to their synthesis. Homeostasis does not belong to this paper anywhere.*

As recommended by the reviewer, we have now removed "homeostasis" from the manuscript and mostly replaced it with "maintenance of cAMP level" in the revised manuscript.

> *The authors have frequently used adverbs at the beginning of a sentence, such as Notably (l.240, 272, 376), Importantly (l.66, 213), More importantly (l.134), Remarkably*

> *(l.264), Interestingly (l.115,301), Intriguingly (l.344), unambiguously (l.347), etc. The use of these words is generally counter-productive. The authors should scan the ms. to eliminate them as far as possible. The sentences would read more clearly and become more impactful.*

Following reviewer's recommendation, we have now eliminated most of the adverbs, mostly used at the beginning of sentences, in the revised manuscript.

> *Specific comments*
>
> *(1) L.1: "maintenance of homeostasis" or increasing cAMP level.*

As suggested by the reviewer, we have now replaced "maintenance of cAMP homeostasis" with "maintenance of cAMP level".

> *(2) L.27: mechanism or reason; varying or various.*

As recommended by the reviewer, we have now replaced "mechanism" with "reason" and the word "varying" is deleted while incorporating suggested changes in the abstract.

> *(3) L.28-29: The logic of connecting PhoP to cAMP doesn't follow well. The logic is much better in l.54, l.112-5 and l.130.*

We thank the reviewer for this suggestion. We have now modified the statement within the 'abstract' in the revised manuscript (duplicated below):

"cAMP is one of the most widely used second messengers which impacts on a wide range of cellular responses in microbial pathogens including M. tuberculosis. Herein, we hypothesized that intra-mycobacterial cAMP level could be controlled by the phoP locus since the major regulator plays a key role in bacterial response against numerous stress conditions."

> *(4) L.30: discovers or reveals (?). Also, in l.101.*

As recommended by the reviewer, we have now replaced 'discovers' with 'reveals' in the Abstract and 'uncovered' with 'revealed' in the Introduction section of the manuscript.

> *(5) L.31: Delete "The most - - derived". It is not obvious what most fundamental means here. I suggest: We find that PhoP-dependent ---involves specific binding of the regulator---PDE gene.*

As recommended by the reviewer, we have modified the statement (duplicated below): "In keeping with these results, we find specific recruitment of the regulator within the promoter region of rv0805 PDE, and absence of phoP or ectopic expression of rv0805 independently accounts for elevated PDE synthesis leading to depletion of intra-mycobacterial cAMP level."

> *(6) L.36: --pathway decreases cAMP level, stress tolerance, and survival of the bacilli.*

As recommended by the reviewer, we have now modified the statement (duplicated below): "Thus, genetic manipulation to inactivate PhoP-Rv0805-cAMP pathway decreases cAMP level, stress tolerance, and intracellular survival of the bacilli.

> *(7) L.41: 'keeps encountering" or encounters?*

As suggested by the reviewer, we have replaced 'keeps encountering' with 'encounters' in the 'Introduction' section of the revised manuscript.

> *(8) L.61: responds, carries.*

Our apologies for the embarrassing grammatical mistakes. We have rectified these errors in the revised manuscript.

> *(9) L.67: you mean burst in synthesis level, not burst of cAMP itself.*

To improve clarity, we have now modified the statement in the revised manuscript (duplicated below): “Agarwal and colleagues had shown that burst in synthesis of bacterial cAMP upon infection of macrophages, improved bacterial survival by interfering with host signalling pathways (Agarwal et al., 2009)”

Reference

Agarwal N, Lamichhane G, Gupta R, Nolan S, Bishai WR (2009) Cyclic AMP intoxication of macrophages by a Mycobacterium tuberculosis adenylate cyclase. Nature 460: 98-102

> *(10) L.77: Change Off to Of.*

We are sorry for the inaccuracy. The suggested change has been made to the text.

> *(11) L.83: Did not discuss "degradation" earlier.*

Following reviewer’s recommendation, we have now modified the statement in the revised manuscript (duplicated below).

“Together, these results strongly suggest that a balance between cAMP synthesis by adenylate cyclases and cAMP degradation by phosphodiesterases contributes to rapid adaptive response of mycobacteria in a hostile intracellular environment (Johnson and McDonough, 2018; McDonough and Rodriguez, 2011).”

Reference

Johnson RM, McDonough KA (2018) Cyclic nucleotide signaling in Mycobacterium tuberculosis: an expanding repertoire. Pathog Dis 76 (5)

McDonough KA, Rodriguez A (2011) The myriad roles of cyclic AMP in microbial pathogens: from signal to sword. Nature reviews Microbiology 10: 27-38

> *(12) L.95: Isn't PhoPR a two-component signal transduction system, the terminology that is more specific than a two-protein regulatory system?*

As recommended by the reviewer, we have replaced “two protein regulatory system” with more specific “two-component signal transduction system” in the revised manuscript.

> *(13) L.124: check-point prevents things from happening. Here the mechanism you found allows growth and survival.*

We agree. As recommended by the reviewer, we have now modified the sentence in the revised manuscript (duplicated below).

“Together, the newly identified mechanism of regulation of cAMP level allows intraphagosomal survival and growth program of mycobacteria.”

> *(14) L.132: why not say directly-"---under normal, and NO and acid stress conditions (Fig. 1A).*

As recommended by the reviewer, we have now deleted the first part of the sentence and directly stated that “we compared cAMP levels………. under normal, NO and acidic stress conditions” (duplicated below).

“We compared cAMP levels of WT and phoPR-KO (lacking both phoP and phoR), grown under normal, NO stress and acid stress conditions (Fig. 1A).”

> *(15) L.134: The complementation is quite variable. Also true in Fig. 2A. If no simple answer, you can say- cAMP values increased in complemented cells, although to a variable extent, for reasons unknown.*

We agree with the reviewer. We have now incorporated new text in the ‘Results’ section of the revised manuscript (duplicated below):

“A higher cAMP level in the complemented strain under NO stress is possibly attributable to reproducibly higher phoP expression in the complemented mutant under specific stress conditions (Khan et al., 2022).”

> *(16) L.154: You rather not say "conclude" and "most likely" at the same time. How about replacing "we conclude" with suggests? In that case, no need to say "most likely". Also, in l.306-7 & l.322-3.*

We thank the reviewer for these suggestions. We have now modified the statements in the revised manuscript (duplicated below).

“We suggest that lower cAMP level of the mutant is not due to its higher efficacy of cAMP secretion.”

Following reviewer’s recommendation, we have incorporated similar changes in two other places of the ‘Results’ section of the revised manuscript.

> *(17) L.161: introduce both the acronyms here and not in l.162.*

Following reviewer’s recommendation, we have made the suggested changes.

> *(18) L.164: Second, (to be in line with First).*

We have made the suggested change.

> *(19). Fig. 2C: There are no black and white bars. This is an important figure because the results appear in the abstract. The signal change from pH 7 to 4.5 is not much. An independent approach would have been desirable. If it were E. coli, I would have suggested beta-gal assay or in vivo footprints. Is a PhoP binding site recognizable in the promoter region of rv0805?*

We apologize for the inaccuracy. We have corrected it in the revised manuscript. Also, we have now carried out DNA binding assays, and included the EMSA data of rv0805 upstream regulatory region binding to phosphorylated PhoP (P~PhoP) as a supplemental figure (Figure 2-figure supplement 1A-B). In this figure, we have also incorporated our results on the likely PhoP binding site within rv0805up. The new figure, figure caption and the relevant results have been adjusted accordingly in the revised manuscript.

> *(20) L.209: ORFs; also delete "of growth" from the sentence.*

The suggested changes were made to the text.

> *(21) L.213: Delete Importantly and change "failed to" to 'did not' (since you did not motivate the expectation earlier, it is better to state the results in an unbiased way).*

As recommended by the reviewer, both changes were included in the revised manuscript.

> *(22) L.217: The requirement of PhoR is a new result - why say "confirm". Change it to indicate. Also, delete "indeed" here and from L.233.*

As recommended by the reviewer, both changes were included in the revised manuscript.

> *(23) L.224: Are the results in Fig 3-S1A under inducing conditions?*

The results shown in Fig 3-S1A are not under inducing conditions of expression. For better clarity, we have modified the sentence describing Figure 3-figure supplement 1A (duplicated below).

"rv0805 ORF was cloned within the multicloning site of integrative pSTki (Parikh et al., 2013) between EcoRI and HindIII sites under the control of Pmyc1tetO promoter, and expression of rv0805 under non-inducing condition was verified by determining the mRNA level (Figure 3 - figure supplement 1A).

Reference:

Parikh et al (2013) Development of a new generation of vectors for gene expression, gene replacement, and protein-protein interaction studies in mycobacteria. Applied and environmental microbiology 79: 1718-1729

> *(24) L.225: ---cAMP level. Add (Fig. 3C) at the end of the next sentence.*

As recommended by the reviewer, both the suggested changes were made to the revised text.

> *(25) L.231: Delete "Most importantly"- you didn't specify what are other less important results.*

We agree. We have now deleted "most importantly" from the sentence in the revised text.

> *(26) L.243 & 254: Change homeostasis to level? Here you are showing mechanisms that can change cAMP level. Homeostasis here would mean how fluctuations in cAMP level are adjusted, usually requiring negative feedback.*

As recommended by the reviewer, 'homeostasis' was replaced with 'level' in both places.

> *(27) L.256: stress response or stress? Also, in l.272*

We are sorry for the inaccuracy. We have corrected these errors in the revised version of the manuscript.

> *(28) L.259: Change "maintenance of homeostasis" to 'repressing the rv0805 PDE gene'. It is safer to use a fact-based title. In this section, direct measurement of rv0805 mRNA, and/or cAMP levels in different genetic backgrounds seem desirable.*

We agree. As recommended by the reviewer, we have modified the title of the 'Results' section in the revised manuscript (duplicated below).

"PhoP contributes to mycobacterial stress tolerance and intracellular survival by repressing the rv0805 PDE expression."

Please note that direct measurements of rv0805 mRNA and cAMP levels are part of Fig. 3 and Figure 3- figure supplement 1A, respectively.

> *(29) Fig, 4A: White and grey symbols are not easily discriminated without zooming. Use color for phoPR-KO.*

We agree. We have now indicated the phoPR-KO in blue in the revised Fig. 4.

> *(30) L.264: Delete remarkable or explain what is so remarkable. Aren't the results expected- the PDE level would go up in both cases. Direct measurement of PDE /cAMP levels would take the mystery out of the results.*

As recommended by the reviewer, we have deleted 'remarkably' in the revised text. We have measured cAMP and PDE expression levels of the four strains in Fig. 3 and Figure 3-figure supplement 1.

> *(31) L.273: --suggesting a role of ---*

We have modified this sentence in the revised version of the manuscript (duplicated below).

"A previous study had reported that phoP-deleted mutant strain was more sensitive to Cumene Hydrogen Peroxide (CHP), suggesting a role of PhoP in regulating mycobacterial stress response to oxidative stress (Walters et al., 2006)."

Reference:

Walters et al. (2006) The Mycobacterium tuberculosis PhoPR two-component system regulates genes essential for virulence and complex lipid biosynthesis. Mol Microbiol 60: 312-330

> *(32) L.275: Delete "transcriptome". CHP sensitivity alone doesn't speak for transcriptome.*

As suggested by the reviewer, we have deleted "transcriptome". Also, please see our response to the previous comment (above).

> *(33) Fig. 4D and E: % Colocalization in the Merge panels is not much different among the four strains tested (to an untrained eye). Can the results be explained to readers not used to in vivo studies?*

As recommended by the reviewer, we have now incorporated new text to explain the in vivo experiment (duplicated below).

"In this assay, WT-H37Rv inhibits phagosome maturation, whereas phagosomes with phoPR-KO mature into phagolysosomes (Anil Kumar et al., 2016)."

Further, for better clarity of the results shown in Fig. 4D, we have (a) increased size of the figure to highlight the difference in the 'merge' panel; (b) included "white arrowheads" in the merge panels of Fig. 4D to indicate auramine labeled mycobacteria, which either have inhibited or facilitated trafficking into lysosomes, and finally (c) incorporated method used to calculate percent co-localization in greater details in the 'Material and Methods' section of the revised manuscript.

Reference

Anil Kumar et al. (2016) EspR-dependent ESAT-6 secretion of Mycobacterium tuberculosis requires the presence of virulence regulator PhoP. J Biol Chem. 291, 19018-19030

> *(34) L.275-6: Delete "next" (also in l.347) and "Note that". In this paragraph, I was expecting some explanation on how phoPR-KO and WT-Rv0805 are behaving similarly. Even if the reason is not known, it should be mentioned.*

The suggested changes have been made to the text. Also, as recommended by the reviewer, we have included the following text in the revised manuscript (duplicated below):

"Together, these results reveal similar behaviour of phoPR-KO, and WT-Rv0805 by demonstrating a comparably higher susceptibility of these strains to acidic pH and oxidative stress relative to WT bacteria and indicate a link between intra-mycobacterial cAMP level and bacterial stress response. Collectively, these data suggest that at least one of the mechanisms by which PhoP contributes to global stress response is attributable to maintenance of cAMP level."

> *(35) L.281: ---WT and indicate a link between cAMP level and stress response in mycobacteria. (No mention of homeostasis).*

The suggested change has been made to the revised text. Please see above our response to point # 34.

> *(36) L.288, 290: No Thus and no clearly.*

Both the suggested changes have been made to the text.

> *(37) L.297: Can you be more direct and state --is due to reduced cAMP level?*

As recommended by the reviewer, we have now modified the sentence to make it more direct in the revised manuscript (duplicated below):

"Together, our findings facilitate an integrated view of our results, suggesting that higher susceptibility of WT-Rv0805 to stress conditions, is attributable to its reduced cAMP level."

> *(38) L.307: May delete "most likely----homeostasis". cAMP is not discussed here. The same deletion is desired in l.324.*

We agree. As recommended by the reviewer, we have now modified the relevant texts in the revised manuscript. These are duplicated below.

"From these results, we suggest that ectopic expression of rv0805 impacts phagosome maturation arguing in favour of a role of PhoP in influencing phagosome-lysosome fusion in macrophages."

"Thus, we suggest that one of the reasons which accounts for an attenuated phenotype of phoPR-KO in both cellular and animal models is attributable to PhoP-dependent repression of rv0805 PDE activity, which controls mycobacterial cAMP level."

> *(39) L.342: cAMP level is regulated remains---*

The suggested change has been made to the revised text (duplicated below):

“Although many bacterial pathogens modulate host cell cAMP level as a common strategy, the mechanism of regulation of mycobacterial cAMP level remains unknown.”

> *(40) L.373: tone down "most fundamental". It is not obvious what is so profound about a stress-response system that depends on PhoP also depends on PhoR. OR justify what is most fundamental about it.*

We agree. Following reviewer’s recommendation, we have modified the text in the revised manuscript (duplicated below):

“In keeping with these results, we find that PhoP-dependent rv0805 expression requires PhoR (Figs. 3A-B), the cognate kinase which activates PhoP in a signal-dependent manner (Gupta et al., 2006; Singh et al., 2023).”

References:

Gupta et al. (2006) Transcriptional autoregulation by Mycobacterium tuberculosis PhoP involves recognition of novel direct repeat sequences in the regulatory region of the promoter. FEBS Letters 580, 5328-5338.

Singh et al. (2023) Dual functioning by the PhoR sensor is a key determinant to Mycobacterium tuberculosis virulence. PLoS Genetics 19(12): e1011070.

> *(41) L.395: delete correspondingly (?)*

The suggested change has been made to the text.

> *(42) L.396: Delete "appear to" and "somewhat". The uncertainty is already implied in "suggest". The evidence that ectopic expression of rv0805 is functionally equivalent to phoP deletion is quite clear in this paper and not saying that clearly is confusing.*

We agree with the reviewer. The suggested changes have been made to the revised text (duplicated below):

“Thus, our results suggest that ectopic expression of rv0805 is functionally equivalent to deletion of the phoP locus.”

> *(43) L.401: --over-expressing bacilli, induction level of rv0805 expression was significantly different in Matange et al and our studies. The next sentence is also very wordy.*

We have made changes to the text to address the reviewer’s concern. Also, the next sentence has been rewritten (duplicated below).

“Although both studies were performed with rv0805 over-expressing bacilli, the fact that important differences in the expression of PDEs, in this study (Matange et al., 2013) and in our assays - yielding significantly different levels of rv0805 expression - most likely account for this discrepancy. While we cannot rule out the possibility of cleavage of other cyclic nucleotides by Rv0805 (Keppetipola & Shuman, 2008; Shenoy et al., 2007; Shenoy et al., 2005), consistent with a previous study our results correlate rv0805 expression with intra-mycobacterial cAMP level (Agarwal et al., 2009).”

References:

Matange et al. (2013) Overexpression of the Rv0805 phosphodiesterase elicits a cAMP-independent transcriptional response. Tuberculosis (Edinb) 93: 492-500.

Keppetipola N, Shuman S (2008) A phosphate-binding histidine of binuclear metallophosphodiesterase enzymes is a determinant of 2',3'-cyclic nucleotide phosphodiesterase activity. J Biol Chem 283: 30942-30949

Shenoy et al. (2007) Structural and biochemical analysis of the Rv0805 cyclic nucleotide phosphodiesterase from Mycobacterium tuberculosis. Journal of molecular biology 365: 211-225

Shenoy et al. (2005) The Rv0805 gene from Mycobacterium tuberculosis encodes a 3',5'-cyclic nucleotide phosphodiesterase: biochemical and mutational analysis. Biochemistry 44: 15695-15704

Agarwal N, Lamichhane G, Gupta R, Nolan S, Bishai WR (2009) Cyclic AMP intoxication of macrophages by a Mycobacterium tuberculosis adenylate cyclase. Nature 460: 98-102

> *(44) L.409: To avoid saying "conclude" and "most likely" at the same time, can you start the sentence thus: 'We infer that Pho-----rv0805 is a---.*

We agree. We have made suggested changes to the text. The modified sentence is duplicated below:

“We infer that PhoP-dependent regulation of Rv0805 is a critical regulator of intra-mycobacterial cAMP level.”

> *(45) L.424. Delete "According to this model". In the preceding sentence, the subject is results, not model. This whole paragraph needs to be rewritten in fewer lines. The shorter the summary statement, the greater would be its impact (less is more here). I would delete the red circles from the figure- it appears that in the repressed state, you are making more products. Replace the circles with an arrow. The legend could be "Increased cAMP level and effective stress response" and "Decreased cAMP---and reduced---.*

We thank the reviewer for these suggestions. Following reviewer’s recommendations, we have made numerous changes and rewritten the paragraph in the revised manuscript (duplicated below):

“In summary, upon sensing low acidic pH as a signal PhoR activates PhoP, P~PhoP binds to rv0805 upstream regulatory region and functions as a specific repressor of Rv0805. Therefore, we observed (a) a reproducibly lower level of cAMP in phoPR-KO relative to WT-H37Rv, (b) a significantly reduced expression of rv0805 in WT-H37Rv, grown under acidic pH relative to normal conditions, and (c) comparable cAMP levels in phoPR-KO and WT-Rv0805. This is why the two strains remain ineffective to mount an appropriate stress response, most likely due to their inability to coordinate regulation of gene expression because of dysregulation of intra-mycobacterial cAMP level. However, without uncoupling regulatory control of PhoPR and rv0805 expression, we cannot confirm that dysregulation of cAMP level accounts for virulence attenuation of phoPR-KO. Given the fact that rv0805-depleted M. tuberculosis is growth attenuated in vivo (McDowell et al., 2023), paradoxically ectopic expression of rv0805 leads to dysregulated metabolic adaptation, thereby resulting in reduced stress tolerance and intracellular survival.”

Also, the suggested changes have been incorporated in Fig. 6 and the figure caption.

Reference

McDowell JR, Bai G, Lasek-Nesselquist E, Eisele LE, Wu Y, Hurteau G, Johnson R, Bai Y, Chen Y, Chan J et al (2023) Mycobacterial phosphodiesterase Rv0805 is a virulence determinant and

its cyclic nucleotide hydrolytic activity is required for propionate detoxification. Mol Microbiol 119: 401-422

> *(46) L.458 & 500: ---was used to transform.*

Following reviewer's recommendation, the suggested changes were made to the text in the Materials and Methods section of the revised manuscript.

> *(47) L.460: --- antibiotics plates.*

Both suggested changes were made to the text.

> *(48) L.466-7: --they were transferred-pH 4.5) and grown for further-*

We thank the reviewer for these suggestions. The suggested changes were made to the text.

> *(49) L.486: ---full-length ORFs of interest were---*

The suggested changes were incorporated in the revised manuscript.

> *(50) L.497: The RNAs were 20 nt long and complementary---*

As recommended by the reviewer, we have modified the text in the revised manuscript (duplicated below).

"The RNAs were 20 nt long and complementary to the non-template strand of the target gene."

> ***Reviewer #2:***
>
> *(1) Rephrase this sentence in the abstract: "Because growing evidence connects PhoP with varying stress response, we hypothesized that the level of 3',5' cAMP, one of the most widely used second messengers, was regulated by the phoP locus, linking numerous stress responses with cAMP production".*

As recommended by the reviewer, we have now rewritten the sentence. The modified text is incorporated in the revised manuscript (duplicated below):

"cAMP is one of the most widely used second messengers, which impacts on a wide range of cellular responses in microbial pathogens including M. tuberculosis. Herein, we hypothesized that intra-mycobacterial cAMP level could be controlled by the phoP locus since the major regulator plays a key role in bacterial responses against numerous stress conditions."

Also, please see our response to specific comments #1-3 of Reviewer 1.

> *(2) Line 134: please describe the complementation strain features as it is mentioned for the first time (plasmid, copy number, promoter etc.) in the manuscript. Especially under NO stress what could be the authors' justification regarding the high cAMP concentration in the complementation strain?*

As recommended by the reviewer, the details of construction of the complemented strain have been incorporated in the 'Materials and Methods' section of the revised manuscript (duplicated below):

"To complement phoPR expression, pSM607 containing a 3.6- kb DNA fragment of M. tuberculosis phoPR including 200-bp phoP promoter region, a hygromycin resistance cassette, attP site and the gene encoding phage L5 integrase, as detailed earlier (Walters et al., 2006) was used to transform phoPR mutant to integrate at the L5 attB site."

To address the reviewer's other concern, we have now included the following sentence in the 'Results' section of the revised manuscript (duplicated below):

"A higher cAMP level in the complemented strain under NO stress is possibly attributable to reproducibly higher phoP expression in the complemented mutant under specific stress condition (Khan et al., 2022)."

Reference:

Khan et al. (2022) Convergence of two global regulators to coordinate expression of essential virulence determinants of Mycobacterium tuberculosis. eLife 2022, 11:e80965.

> *(3) In Figure 1C, it is a bit confusing to see the numbers 1,2,3 and 4 and nothing is referred to these numbers in the figure legend so it's better to remove them.*

We agree with the reviewer. We have now removed the lane numbers from the figure (Fig. 1C) in the revised manuscript.

> *(4) Line 852: rephrase it "insignificantly different".*

The suggested change has been made to the text. The modified text is incorporated in the manuscript (duplicated below):

"Note that the difference in expression levels of rv0805 between WT and phoPR-KO was significant ($p<0.01$), whereas the fold difference in mRNA level between WT and the complemented mutant (Compl.) remains nonsignificant (not indicated)."

> *(5) Line198-200: There are no open/black bars, they all are coloured bars. Correct the same. The significance test should be done for the same gene (suppose rv0805 up) in different pH conditions. Right now, it is not revealing anything and misleading.*

We apologize for the inaccuracy. We have now rectified the error. As recommended by the reviewer, Fig. 4C was modified, and the significance tests were carried out between samples involving identical promoter enrichments under different pH conditions. The modified figure, figure legend, and the relevant results have been adjusted accordingly in the revised manuscript.

> *(6) Line 213: Is there any difference between this complementation strain (phoPR-KO:: phoPphoR with the one used in Figure 1A, 1B, and 2A? If yes, then please describe it.*

The same complemented mutant strain, which has been described in the 'Materials and Methods' section of the revised manuscript, was used in the experiments described in Fig. 1A, Fig.1B and Fig. 2A.

> *(7) Line 223: Please mention the copy number and promoter of the vector construct.*

As recommended by the reviewer, we have now mentioned the promoter of the vector and incorporated new text with regard to copy number of the expression vector in the revised manuscript (duplicated below).

"Although copy number of episomal vectors with pAl5000 origin of replication (oriM) have been reported to be 3 by Southern hybridization (Ranes et al, 1990), in this case wild-type and mutant Rv0805 proteins were expressed from single-copy chromosomal integrants (Parikh et al., 2013).”

References

Ranes et al., (1990) Functional analysis of pAL5000, a plasmid from Mycobacterium fortuitum: construction of a "mini" mycobacterium-Escherichia coli shuttle vector. J Bacteriol 172: 2793-2797

Parikh et al., (2013) Development of a new generation of vectors for gene expression, gene replacement, and protein-protein interaction studies in mycobacteria. Applied and environmental microbiology 79: 1718-1729

> *(8) Figure 3 - Figure Supplement 1: not sure why the authors measured mRNA levels of rv1357 and rv2387? These genes were not overexpressed!*

The mRNA levels of rv1357 and rv2387 were measured to show that overexpression of either the wild-type or mutant Rv0805 did not influence expression of other PDEs like Rv1357 and Rv2387. We have now mentioned it explicitly in the revised manuscript (duplicated below).

“In contrast, other PDE encoding genes (rv1357 and rv2387), under identical conditions, demonstrate comparable expression levels in WT-H37Rv and rv0805 over-expressing strains.”

> *(9) Line 234: Wrong interpretation it should be PDE mRNA levels in WT-Rv0805 and WT-Rv0805M.*

As recommended by the reviewer, we have now modified the statement to improve clarity (duplicated below).

“The corresponding mRNA levels of PDEs (wild-type and the mutant) are over-expressed approximately 4.5-6 -fold relative to the genomic rv0805 level of WT-H37Rv (Figure 3-figure supplement 1A).”

> *(10) Line 237: Remove the sentence "Thus, we conclude......identical expression strategy", you have already talked about why phosphodiesterase activity is crucial for cAMP concentration and it is well understood.*

Following reviewer’s recommendation, we have now removed the sentence from the revised manuscript.

> *(11) Figure 3E: Authors should comment on why the cAMP concentration is not significantly changed even though the mRNA level changes are drastic (~90%). How do you correlate that? Is it because of other PDEs?*

We agree. As suggested by the reviewer, we have now incorporated new text in the revised manuscript (duplicated below).

“We speculate that effective knocking down of phoP or rv0805 is not truly reflected in the extent of variation of cAMP levels possibly due to the presence of numerous other mycobacterial PDEs.”

> *(12) Line 505,506: Is it the translation start site or the transcription start site? Because mRNA level changes are reported.*

It is the translational start sites, and gene-specific small guide RNAs were designed to inhibit mRNA expression.

> *(13) Line 292: There is a difference between red and green bars. Authors should do statistical analysis and then comment on whether overexpression of WT and mutant pde are different or similar, to me they are different; also, explain why the WT-Rv0805 strain is different than the phoPR-KO strain in the context of cell wall metabolism.*

As recommended by the reviewer, we have now included statistical significance of the data in the revised version, and modified the text accordingly in the manuscript.

Also, we included text explaining why WT-Rv0805 is different compared to phoPR-KO strain in the context of cell wall metabolism (duplicated below).

"Together, these results suggest that both strains expressing wild type or mutant PDEs share a largely similar cell-wall properties and are consistent with (a) a recent study reporting no significant effect of cAMP dysregulation on mycobacterial cell wall structure/permeability (Wong et al., 2023), and (b) role of PhoP in cell wall composition and complex lipid biosynthesis (Walters et al., 2006; Asensio et al., 2006; Goyal et al., 2011)."

References:

Wong et al. (2023) Cyclic AMP is a critical mediator of intrinsic drug resistance and fatty acid metabolism in M. tuberculosis. eLife 2023; 12: e81177

Walters et al. (2006) The Mycobacterium tuberculosis PhoPR two-component system regulates genes essential for virulence and complex lipid biosynthesis. Mol Microbiol 60: 312-330

Asensio et al. (2006) The Virulence-associated Two-component PhoP-PhoR System Controls the Biosynthesis of Polyketide-derived Lipids in Mycobacterium tuberculosis. J Biol Chem 281: 1313-1316.

Goyal et al. (2011) Phosphorylation of PhoP protein plays direct regulatory role in lipid biosynthesis of Mycobacterium tuberculosis. J Biol Chem 286: 45197-45208

> *(14) Line 299-303: Authors should explain how the colocalization % are calculated. Also, in the figure 4D merge panel please highlight the difference.*

As suggested by the reviewer, we have now explained the methodology used to calculate percent colocalization in greater details. Also, we have modified Figure 4D to highlight the difference between samples shown in merge panel. Please see our response to comment # 33 from the Reviewer 1.

> *(15) General comment: There are multiple instances where writing needs to be improved.*

We are sorry for the inaccuracies. We have now done thorough editing of the manuscript and made numerous corrections throughout.